
Holographic Flow Networks

Quoc Binh Ta



LMU Munich 2022

Contents

1	Introduction	1
2	AdS/CFT Correspondence	5
2.1	Conformal two-point function	6
2.2	Boundary values as sources	8
2.3	Holographic renormalization	10
3	Normalizing Flows	15
3.1	Tractable probability density	16
3.2	Unitary flows	18
3.3	Kullback-Leibler divergence	19
3.4	Reparametrization trick	21
4	Holographic RG with Flow Networks	23
4.1	General architecture	24
4.2	Discrete field theories	28
4.3	Flow design	35
4.4	Training and results	40
4.5	Model extensions	47
5	Outlook	51
A	Code Example	59

1 Introduction

Over the last decade, tremendous progress has been made in the field of artificial intelligence (AI). From self-driving cars to protein folding: machine learning (ML) has shown itself to be inevitable for future technologies. But not only has it transformed the industry world, it has also become an integral part of scientific research. The processing and analysis of huge amounts of data, which are typical for experiments in high energy or astrophysics, pose major challenges to traditional methods. With the rise of AI, we can tackle these tasks in an efficient way. Although the questions, which are addressed in theoretical physics, usually do not involve a lot of data, they can be equally or even more difficult to answer. Solving them, however, is of great importance because a deep understanding of nature is very often the basis for technological inventions.

One such topic, which has attracted a lot of interest, is the holographic principle. It states that quantum gravity (or string theory) can be encoded on the boundary of the spacetime on which it is defined on. The theory therefore can be thought of as being a holographic projection of the information onto a lower dimensional space. It turns out that this defines a dual non-gravitational theory on the boundary. Two physical theories without any apparent connection can somehow ultimately describe the same physics. This is also called the gauge/gravity duality, because the boundary system is often found to be a QFT with gauge symmetry. The gravity theory is related to it in a special way: whenever one theory is in the strong coupling regime, the other is weakly coupled. This opens up the possibility to analyze a system at energy scales where perturbation theory would usually break down. To do that we have to find its holographic dual. If this relationship between the two descriptions is actually true, then information about gravitational systems or even quantum gravity itself can be accessed. An example for such systems are black holes. As a matter of fact, the holographic principle was inspired from a result about the their entropy, which states that the maximal information contained in black holes scales with their boundary

surface [7]. On a more practical side, the duality could enable us to investigate strong coupling phenomena of the gauge theories like holographic QCD. Although this theory is not the same as the physical QCD with the $SU(3)$ color gauge group, it could still give us insights about aspects common to QCDs. An important concept is for example the confinement of quarks, which is still puzzling to this day. An understanding of its underlying mechanism is tightly intertwined to the question of proton stability. With the holographic principle we might be able to tackle these difficult tasks. Unfortunately, it is not clear how to find a holographic dual and how to precisely map the two theories to each other.

Among the various formulations and versions of holography the most popular duality is arguably the AdS/CFT correspondence. Its general statement is that any conformal field theory (CFT) defined a conformally flat spacetime of d dimensions is equivalent a quantum gravity theory in a bulk spacetime, which is asymptotically $AdS_{d+1} \times C$, with C being some compact manifold. Although this correspondence has been analyzed in great detail over the last two decades, it still remains unproven and the equivalence between gauge theory and gravity is only conjectured. Could machine learning perhaps help us gain a deeper understanding of the correspondence?

Since the appearance of neural networks many architectures with impressive results have been introduced. Of particular interest are the generative models, which can learn a (sometimes even unknown) probability distribution and create samples from it. Variational autoencoders (VAEs), Boltzmann machines, generative adversarial networks (GANs) or normalizing flows are some examples for those networks. The ability to sample efficiently from a learned distribution is especially important when the original probability density cannot be computed due to its complexity. This is the case for quantum field theories where the partition function can only be calculated exactly for Gaussian distributions. In this situation, normalizing flows can help us. Using this type of neural network to approximate the probability distribution does not only lead to efficient sampling but also a tractable density, i.e. the learned distribution can be directly accessed. This is one of the reasons, why flow models will be our architecture of choice.

In order to apply such a network to holography, we have to make use of a peculiar fact about the interior of the bulk space. As mentioned above, the CFT is defined on the AdS boundary. But when we move from the boundary into the bulk space

along the radial (or holographic) direction, which is perpendicular to the boundary surface, we can find another CFT. This new theory turns out to be a renormalized version of the original one. The flow into the bulk interior corresponds to an RG-flow of the CFT and the radial coordinate can be identified with the inverse RG-scale. If we can set up a neural network, which can perform an RG-flow on a field theory, then we might find an emergent geometric structure, corresponding to a hyperbolic space, in the network.

In the following we will first study the AdS/CFT correspondence in more detail to gain a better understanding of the physical framework. After that, the flow networks are reviewed alongside with features to be used later. We are then ready to build an holographic flow network and apply it to a discrete QFT on a lattice. The training of various networks is discussed and methods to extract geometrical information from the flow are presented. At last we will mention some applications for the trained flow and show possible extensions of the model.

2 AdS/CFT Correspondence

A first concrete example of the holographic principle was given in [22], where it was conjectured the equivalence between a $\mathcal{N} = 4$ supersymmetric $SU(N)$ Yang-Mills theory with conformal symmetry and type IIB string theory on $AdS_5 \times S^5$. This was motivated by the fact, that by taking specific limits of type IIB string theory on a background curved by N coincident D3-branes one can obtain either one of the two descriptions (see [10] for a detailed review). But the equivalence is already apparent at the level of symmetry. Since the two theories are conjectured to be the same, their symmetry groups have to match. In general, the conformal field theory lives in a d -dimensional flat spacetime and the gravity theory is defined in a $(d+1)$ -dimensional hyperbolic background, which is usually defined as a hypersurface of a $(d+2)$ -dimensional flat space. This means that an AdS_5 space inherits the Lorentz isometry $SO(4,2)$ from its embedding, which has 15 generators. The symmetries of a CFT on the other hand are generated by the usual 10 Lorentz generators corresponding to translations, rotations and boosts with additional 5 conformal symmetries, which include four special conformal and one scale transformation. These 15 generators satisfy the same algebra as the isometry group of the AdS space. In fact, $SO(4,2)$ is also called the conformal group. On the gravity side we have an additional $SO(6)$ symmetry from the compactification on the sphere S^5 . This matches to the $\mathcal{N} = 4$ SUSY of the field theory, which has symmetry group $SU(4)$, since the two groups are isomorphic to each other: $SO(6) \cong SU(4)$.

To see that the CFT actually lives on the boundary of the bulk space, we look at the metric of AdS_{d+1} in global coordinates [6], which is given by

$$ds^2 = \frac{L^2}{\cos^2 \rho} (-d\tau^2 + d\rho^2 + \sin^2 \rho d\Omega^2) , \quad (2.1)$$

with $-\pi < \tau \leq \pi$, $0 \leq \rho < \frac{\pi}{2}$ and $\sum_{i=1}^d \Omega_i^2 = 1$. We observe that the boundary is the hypersurface at $\rho = \frac{\pi}{2}$ and has the topology of $\mathbb{R} \times S^{d-1}$, which is conformal

to d -dimensional Minkowski space. Therefore, a conformally invariant QFT can be thought of as living on the conformal boundary of AdS_{d+1} . The field theory contains the same amount of information as the gravity theory, or in other words, the bulk system can be holographically projected and encoded onto the boundary surface. With a change of coordinates and a conformal rescaling we can bring the metric (2.1) into the simple form

$$ds^2 = \frac{L^2}{z^2} (dz^2 + \eta_{ij} dx^i dx^j) , \quad (2.2)$$

which is referred to as Poincaré coordinates and covers half of AdS_{d+1} . This is most common and convenient metric for hyperbolic spaces. Its boundary lies at $z = 0$ with Minkowski metric η_{ij} .

For easier calculations and implementations we will choose to use the Euclidean version of AdS space, the hyperbolic space $\mathbb{H}^{d+1}$. We will however continue to refer to this space as AdS, since a $\mathbb{H}^{d+1}$ metric can be easily obtained from the Minkowski version by Wick rotation to imaginary time. With this, the boundary metric becomes $\eta_{ij} = \delta_{ij}$. Bulk indices will be denoted by greek letters μ, ν and boundary indices by Latin letters i, j . We will use $x, y \in \mathbb{R}^d$ as coordinates of the boundary with $x = (x^0, \dots, x^d)$ and $z \in \mathbb{R}$ as the holographic radial direction.

To analyze the correspondence, we would like to find some observables which can be computed in both theories. By comparing results, we can then give us a first check, whether the holographic principle is realized at all. A simple quantity to calculate is the two-point function or correlator.

2.1 Conformal two-point function

Due to the large group of Poincaré and conformal transformations, the general form of the one, two and three-point functions are completely determined by symmetry considerations up to constant factors. Suppose we have two complex fields $\psi_1(x), \psi_2(y)$ with conformal dimensions Δ_1, Δ_2 on the boundary. Because of translation invariance, we already know that the correlator between the two fields has to be a function of the distance, i.e.

$$\langle \psi_1(x) \psi_2^*(y) \rangle = f(|x - y|) . \quad (2.3)$$

In a CFT we are allowed to perform scaling transformations $x \rightarrow \lambda x$ without changing the observables. A field will transform under scaling like

$$\psi_i(\lambda x) = \lambda^{\Delta_i} \psi_i(x) . \quad (2.4)$$

With this we can restrict the form of the two-point function by imposing

$$\begin{aligned} \langle \lambda^{\Delta_1} \psi_1(\lambda x) \lambda^{\Delta_2} \psi_2^*(\lambda y) \rangle &= \lambda^{\Delta_1 + \Delta_2} f(\lambda(x - y)) \\ &\stackrel{!}{=} f(x - y) \end{aligned} , \quad (2.5)$$

which is satisfied when f decays according to a power-law in the distance $|x - y|$ like

$$f(|x - y|) \sim \frac{1}{|x - y|^{\Delta_1 + \Delta_2}} . \quad (2.6)$$

A relation between the conformal dimensions Δ_1 and Δ_2 can be found by considering inversions of the form

$$x \rightarrow \frac{x}{|x|^2} . \quad (2.7)$$

After performing this transformation on the correlator and imposing the invariance under them, we find the condition:

$$\begin{aligned} \left\langle \frac{1}{x^{2\Delta_1} y^{2\Delta_2}} \psi_1\left(-\frac{1}{x}\right) \psi_2^*\left(-\frac{1}{y}\right) \right\rangle &= \frac{1}{x^{2\Delta_1} y^{2\Delta_2}} \frac{c}{\left(-\frac{1}{x} + \frac{1}{y}\right)^{\Delta_1 + \Delta_2}} \\ &\stackrel{!}{=} \frac{c}{|x - y|^{\Delta_1 + \Delta_2}} . \end{aligned} \quad (2.8)$$

To fulfill this constraint, we have to set $\Delta_1 = \Delta_2$. This means that the correlator between two fields is only non-zero, if their conformal dimensions are the same. So in general, the two-point function in a CFT will always be an algebraic decay

$$\langle \psi(x) \psi^*(y) \rangle \sim \frac{1}{|x - y|^{2\Delta}} . \quad (2.9)$$

This gives us a practical way to check if a theory is conformal. Such a power-law usually implies the presence of an infinite correlation length.

2.2 Boundary values as sources

The two-point function of a CFT can also be calculated using the bulk gravity action, which establishes a concrete connection between the two theories. We will follow the original derivation of [28]. Assume that there exist a term in the field theory action which couples a CFT observable $\mathcal{O}$ with an external source J . In general, the interaction between them has the form

$$\int dx \mathcal{O}(x) J(x) . \quad (2.10)$$

The usual procedure for a QFT is to construct a partition function $Z[J]$ and calculate the n-point functions as functional derivatives of $Z[J]$ with respect to the external source at $J = 0$. In the following, we will express the bulk action in terms of boundary values of the classical bulk fields.

Consider a bulk theory with a free complex scalar field $\phi(x, z)$ in a general background with action

$$S_{\text{bulk}}[\phi, g_{\mu\nu}] = \int d^d x dz \sqrt{g} \left[g^{\mu\nu} \partial_\mu \phi^* \partial_\nu \phi + m^2 |\phi|^2 + R \right] . \quad (2.11)$$

We fix the geometry to be AdS_{d+1} and find the classical equation of motion for ϕ by using integration by parts. For the ϕ -dependent terms we get

$$S_{\text{bulk}}[\phi] = - \int d^d x dz \sqrt{g} \phi^* (-\square_g + m^2) \phi + \int d^d x dz \partial_\mu (\sqrt{g} g^{\mu\nu} \phi^* \partial_\nu \phi) \quad (2.12)$$

with the Laplacian

$$\begin{aligned} \square_g &= \frac{1}{\sqrt{g}} \partial_\mu (\sqrt{g} g^{\mu\nu} \partial_\nu) \\ &= \frac{z^2}{L^2} (\partial_z^2 + (1-d) \frac{1}{z} \partial_z + \partial_x^2) . \end{aligned} \quad (2.13)$$

The second term in equation (2.12) does not contribute to the e.o.m. because it can be written as a surface integral by invoking Stokes' theorem. After imposing $\delta S = 0$, we arrive at

$$z^2 \partial_z^2 \phi - (d-1) z \partial_z \phi + (z^2 \partial_x^2 - L^2 m^2) \phi = 0 , \quad (2.14)$$

where the general solution for will be a convolution of some initial configuration ϕ_0

and the Green's function K , satisfying

$$(-\square_g + m^2)K(x, z) = \delta(x, z) . \quad (2.15)$$

To solve this, we look for a solution of the homogeneous equation which has a singularity at one point on the boundary. This corresponds to the behavior we would expect for a δ -function. Since we have translation invariance in x direction, the Green's function should only be a function of z . As an ansatz we take

$$K(z) = z^\alpha . \quad (2.16)$$

By changing coordinates using the inversion map $x \rightarrow \frac{x}{z^2 + |x|^2}$, $K(z)$ becomes

$$K(z) = \frac{z^\alpha}{(z^2 + |x|^2)^\alpha} . \quad (2.17)$$

This function has a singularity at $x = 0$ for $z \rightarrow 0$, which means that it behaves like a δ -function on the boundary. The field ϕ can now be expressed by a convolution of the form

$$\phi(x, z) \sim \int dy K(x - y, z) \phi_0(y) , \quad (2.18)$$

where $\phi_0(x) \equiv \phi(x, z = 0)$. We can use this expression to write the action as a functional of the boundary values ϕ_0 . The second term in the action (2.12) is a surface integral

$$\int d^d x \sqrt{h} \phi^* n^\mu \partial_\mu \phi = \int d^d x \frac{1}{z^{d-1}} \phi^* \partial_z \phi \quad (2.19)$$

with induced metric h_{ij} and normal vector n^μ . The derivative of ϕ with respect to z can be estimated to lowest order in z from the classical solution (2.18) as

$$\partial_z \phi \sim z^{d-1} \int dy \frac{\phi_0(y)}{|x - y|^{2\alpha}} . \quad (2.20)$$

With this, the bulk action on the boundary will be of the form

$$S_{\text{bulk}}[\phi_0] = \int dx dy \frac{\phi_0^*(x) \phi_0(y)}{|x - y|^{2\alpha}} , \quad (2.21)$$

from which the CFT two-point function can be calculated.

Assume, that a complex boundary field $\psi(x)$ and the external source $\phi_0(x)$ have an

interaction of the form

$$- \int dx [\psi(x)\phi_0^*(x) + \psi^*(x)\phi_0(x)] . \quad (2.22)$$

Since the AdS/CFT correspondence states that the boundary and bulk theories are the equivalent, their partition functions should contain the same information, i.e.

$$Z_{\text{CFT}} = Z_{\text{bulk}} \approx e^{-S_{\text{bulk}}[\phi_0]} . \quad (2.23)$$

In the saddle point approximation of the gravity theory, the classical bulk action plays the role of a generating functional for the field theory and we can compute the two-point function for ψ by taking functional derivatives of Z_{bulk} with respect to the sources ϕ_0 :

$$\begin{aligned} \langle \psi^*(x)\psi(y) \rangle &= \frac{\delta}{\delta \phi_0(x)} \frac{\delta}{\delta \phi_0^*(y)} Z_{\text{bulk}}[\phi_0] \\ &\sim \frac{1}{|x-y|^{2\alpha}} . \end{aligned} \quad (2.24)$$

Thus, we have found a polynomial decay of the correlation function, which is consistent with the result from the CFT. The exponent can be identified with the conformal dimension $\Delta = \alpha$. We have obtained way to calculate any n-point function of the QFT without precise knowledge of its action or dynamics, since the information is provided by the bulk theory. One should note that this is the weakest form of the AdS/CFT correspondence because it is only valid in the saddle point approximation of the bulk theory, e.g. the supergravity limit of string theory. In the case of the original correspondence involving SUSY Yang-Mills theory with gauge group $SU(N)$, this also implies the large N limit $N \rightarrow \infty$. However, the equivalence is believed to hold even in the quantum regime and on all energy scales.

2.3 Holographic renormalization

When calculating observables of a QFT, one usually has to do them carefully in a renormalized fashion, as quantities can easily diverge due to the presence of arbitrary small scales or infinite volumes in the definition of the bare action. A potential divergence is, for example, the integration $\int d^d x$ over the boundary at $z = 0$, since the area is infinite. Before we perform any calculations, we have to regularize the

theory by introducing a cutoff to the length or energy scale, e.g. area integral goes over a surface at some $z_0 \neq 0$. We can send $z \rightarrow 0$ after the calculation to remove the cutoff. This transformation is part of the isometry $(x, z) \rightarrow (\lambda x, \lambda z)$ of AdS space which is dual to the dilation symmetry of CFTs. Regarding the boundary theory, this process corresponds to a coarse-graining which induces an RG-flow.

As an example let's consider again a massive field $\phi(x, z)$ in the bulk. A typical correlation function between two points in at $z = z_0$ will decay exponentially with the lengths of paths connecting them [13] as

$$\langle \phi(x) \phi^*(y) \rangle_{z=z_0} \sim \sum_{\text{paths}(x,y)} e^{-m \mathcal{D}(x,y)} . \quad (2.25)$$

This expression will be dominated by the geodesic between the points (x, z_0) and (y, z_0) for $m \gg L^{-1}$, such that

$$\langle \phi(x) \phi^*(y) \rangle_{z=z_0} \sim e^{-m \mathcal{D}_{\text{geo}}(x,y)} . \quad (2.26)$$

The geodesic in AdS space will be of logarithmic dependence on $|x - y|$ and is given by

$$\mathcal{D}_{\text{geo}}(x, y) = L \ln \left(\frac{\left(|x - y| + \sqrt{|x - y|^2 + z_0^2} \right)^2}{z_0^2} \right) . \quad (2.27)$$

For large distances $|x - y| \gg z_0$, this can be approximated as

$$\mathcal{D}_{\text{geo}}(x, y) \approx L \ln |x - y| . \quad (2.28)$$

We now see that the two-point function in a hyperbolic space will be roughly a power-law given by

$$\langle \phi(x) \phi^*(y) \rangle_{z=z_0} \sim \frac{1}{|x - y|^{2mL}} . \quad (2.29)$$

This is the familiar form, that we get for a CFT, with identification $\Delta = mL$. However, this functional dependence is only approximate and the precise behavior at short scales is not polynomial and therefore modified. This implies that the long distance physics of the bulk theory is related to the UV dynamics of the boundary. This is called the IR-UV relation or duality which was first discussed in [26]. The effective theory at z_0 is a renormalized CFT, due to the UV-cutoff induced by the

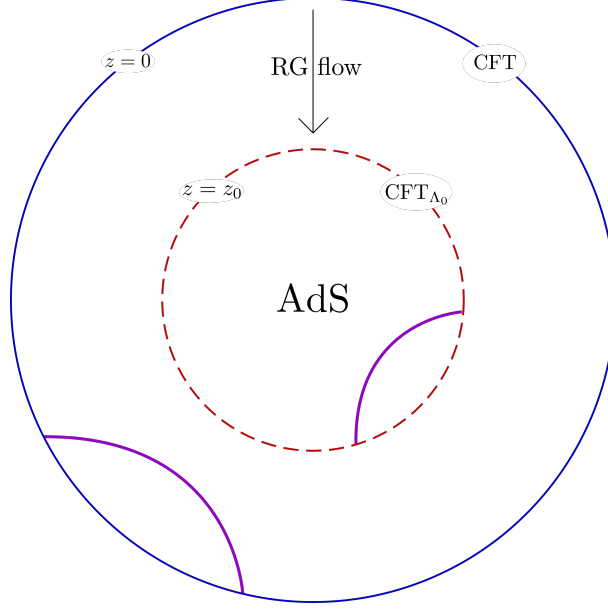


Figure 2.1: The flow in the radial direction z of AdS space is equivalent to an RG-flow of the CFT. The violet lines represent bulk geodesics.

radial movement into the bulk. The energy scale of the RG-flow is

$$\Lambda_0 = \frac{1}{z_0} . \quad (2.30)$$

This implies a relation of the renormalization group flow of the CFT with the radial flow of the gravity theory, see figure 2.1. The information of the interior space enclosed by the dotted line is encoded on its surface area. The Bekenstein bound for the entropy of this smaller spacetime volume is satisfied. One can make this relationship more concrete. But as with the calculation of the two-point function, a precise computation is only possible in the saddle point approximation.

In a Wilsonian approach, we would integrate out the UV degrees of freedom in the CFT, which should correspond to an integration over the long wavelength modes in the bulk [17]. In the classical limit, we can then define an inner correspondence between the CFT at z_0 and the interior volume of the AdS space by integrating out bulk fields in the excluded area for $z < z_0$ [4].

A even more precise mapping can be achieved when we go into the Hamilton-Jacobi formulation of the gravity theory, where the first order flow equations can be cast into the form of Callan-Symanzik equations [8]. In a semiclassical treatment, the field values $\phi(x, z)$ can be interpreted as the running coupling constants of the CFT.

The Hamilton equations of motion for the bulk fields are then equivalent to the renormalization group flow equations of the CFT [3] and we can map

$$z \frac{d}{dz} \phi = \frac{\delta \mathcal{H}}{\delta \pi} \leftrightarrow \Lambda \frac{d}{d\Lambda} g = \beta(g) , \quad (2.31)$$

where $\mathcal{H}$ is the Hamiltonian, π the momentum variable conjugated to ϕ , Λ the energy scale of the RG-flow with a running coupling constant g . The metric of the bulk is therefore a metric of the space of RG-flows and the value of the field ϕ at different radial positions is the running coupling constant. However, such a mapping is only valid in the semiclassical limit and also depends on the renormalization scheme. This means that one has to take precautions when setting up a coarse-graining procedure for a CFT, if we want the RG-flow space to be described a metric. Because there exist various renormalization schemes, not all RG-flow spaces will exhibit a geometric structure even if there exist a dual gravity description for the CFT.

3 Normalizing Flows

An important goal in statistical modeling is the sampling from a desired target probability distribution $q_{\text{target}}(x)$. Unfortunately, very often we are not able to do so directly because the probability density of interest is not tractable. This is usually due to high dimensional integrals, which appear in the calculation of the normalization constant. To circumvent this problem, we can make use of Monte Carlo methods like the Metropolis-Hastings algorithm, where knowing a function proportional to q_{target} is sufficient. However, Monte Carlo sampling can be computationally quite expensive. A more efficient approach is to utilize generative ML models. One of the first network structures with the ability to capture a probability density are Boltzmann machines. The binary nodes of this type of memory model, the so-called Hopfield network, behave exactly the same way as spins in an Ising model. This means that by choosing the energy of the Ising model as the loss function, we are effectively searching for an equilibrium state during the network training. The configuration of the units will then follow the Boltzmann distribution $p \sim e^{-E}$. Although samples from this density can be easily obtained by taking random configurations of the nodes, the underlying distribution function is not accessible. This drawback is also present in other more efficient ML models like autoencoders or GANs.

A solution is provided by normalizing flows [25]. These neural networks can evaluate the learned distribution exactly because it is obtained by the application of a series of invertible transformations on a simple base distribution, typically taken to be a Gaussian. The flow network effectively performs a change of variables on normal distributions. Since it is invertible, the reverse flow map can also be used. Training the network then equates to finding a set of maps, which transform a complicated probability density into the product of normal distributions, i.e. the network normalizes the target distribution. A detailed review of normalizing flows is [23], which we will largely follow.

We will present some basic concepts for flow networks in the case of complex vari-

ables. This is a bit unusual because neural networks are usually build with real variables, where complex numbers are implemented as 2D vectors. But there has been great effort recently to implement complex tensors natively into the standard ML-libraries.

In this chapter we will denote the data vector by $x = (x^1, \dots, x^N)^T \in \mathbb{C}^N$ and the (latent) variable of the base distribution by $z = (z^1, \dots, z^N)^T \in \mathbb{C}^N$, which is common in the ML literature.

3.1 Tractable probability density

In the ideal case, the network transformation G^{-1} maps the input data x , which follows the target distribution $q_{\text{target}}(x)$, to the latent random variables $G^{-1}(x) = z \sim p_{\text{base}}(z)$. Conversely, we can also sample from the base distribution $p_{\text{base}}(z)$ by applying the inverse map on z to obtain new data $x = G(z)$. The two densities are related by a change of variables with the Jacobian matrix

$$J_G(z) = \begin{pmatrix} \frac{\partial G^1}{\partial z^1} & \cdots & \frac{\partial G^1}{\partial z^N} \\ \vdots & \ddots & \vdots \\ \frac{\partial G^N}{\partial z^1} & \cdots & \frac{\partial G^N}{\partial z^N} \end{pmatrix}. \quad (3.1)$$

Because we are using complex numbers and $\mathbb{C} \cong \mathbb{R}^2$, the vector x actually represents $2N$ independent variables. So when we change from x to z , we have to account for contributions coming from x and also its conjugate $x^\dagger$. If our flow treats the real and imaginary part of x and z in the same way, i.e. they transform under the same map, then $\det J_{G^{-1}}^*(x) = \det J_{G^{-1}}(x^\dagger)$ and the target density can then be rewritten as

$$\begin{aligned} p_{\text{flow}}(x) &= p_{\text{base}}(G^{-1}(x)) \left| \det \frac{\partial G^{-1}}{\partial x} \right| \left| \det \frac{\partial G^{-1}}{\partial x^\dagger} \right| \\ &= p_{\text{base}}(z) \left| \det \frac{\partial G}{\partial z} \right|^{-1} \left| \det \frac{\partial G}{\partial z^\dagger} \right|^{-1} \\ &= p_{\text{base}}(z) |\det J_G(z)|^{-2}, \end{aligned} \quad (3.2)$$

where the switch from G^{-1} to G is allowed by the inverse function theorem. We can choose almost any density as the base, even a very restricted one like the uniform distribution. In fact, if $q_{\text{target}}(x)$ and $p_{\text{base}}(z)$ are well-behaved, then a diffeomorphism transforming between base and target density is guaranteed to exist (see [23])

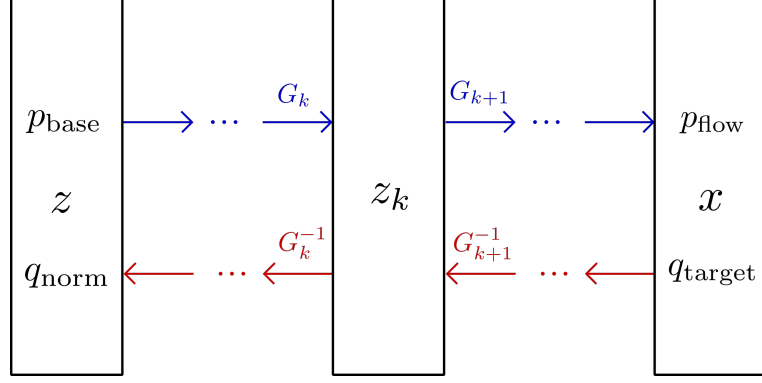


Figure 3.1: Flow network consisting of multiple bijective maps

for a proof).

The extension to multiple bijective maps G_k is straightforward and can be done by simple composition of the transformations, see figure 3.1. The full map of the flow consisting of K invertible layers is then given by

$$G = G_K \circ \dots \circ G_1 \quad (3.3)$$

with inverse transformation

$$G^{-1} = G_1^{-1} \circ \dots \circ G_K^{-1} . \quad (3.4)$$

The Jacobian determinant for G is still easily accessible as a product of each layer:

$$\begin{aligned} \det J_G(z) &= \det J_{G_K}(G_{K-1}(z_{K-2})) \cdots \det J_{G_1}(z) \\ &= \prod_{k=1}^K \det J_{G_k}(z_{k-1}) , \end{aligned} \quad (3.5)$$

where we defined $z_k = G_k(z_{k-1})$ and identified $z_0 \equiv z$, $z_K \equiv x$. By collecting the Jacobian at each layer during a pass through the flow network, one can easily obtain the determinant of the full transformation. Very often, we will not directly handle the determinants, but instead calculate the logarithm of them. This is because in practice one usually works with log probabilities, as they are additive and therefore easier to manipulate. This is also favorable from the computational point of view, because determinants can easily blow up in value for high dimensional data. Calculating the logarithm of Jacobians directly is more stable. For the expression (3.5)

we get

$$\log |\det J_G(z)| = \sum_{k=1}^K \log |\det J_{G_k}(z_{k-1})| , \quad (3.6)$$

which is now a summation over the layer contributions instead of a product.

3.2 Unitary flows

The process of designing the transformation G_k is usually guided by the premises to have tractable Jacobians and inverse maps. Ideally, the log determinants are easy to compute and the inverse maps are of simple form. But the only necessary requirement is the existence of the inverse or a non-zero Jacobian. One could implement G_k as entire neural networks, which is the case for flows like RNVPs. In the simplest case, G_k is just a linear layer. By constraining the weight matrix to be orthogonal or unitary, we get a trivial Jacobian and $\log |\det J| = 0$. The computation of the inverse for unitary matrices also requires no special solving algorithm since only hermitian conjugation is needed. We will discuss two ways to transform a weight matrix $\tilde{W}$ into a unitary layer U .

Assume that we have a upper triangular matrix with zeros on the diagonal of the form

$$\tilde{W} = \begin{pmatrix} 0 & w_{1,2} & \dots & w_{1,n} \\ 0 & 0 & \ddots & \vdots \\ \vdots & & \ddots & w_{n-1,n} \\ 0 & 0 & \dots & 0 \end{pmatrix} . \quad (3.7)$$

$\tilde{W}$ can be made into a skew-hermitian (or anti-hermitian) matrix by

$$W = \tilde{W} - \tilde{W}^\dagger . \quad (3.8)$$

Since the space of skew-hermitian matrices with zero trace form the Lie algebra of the special unitary group $SU(n)$, we can apply the exponential map on W and get a unitary layer U for the flow:

$$U = e^W . \quad (3.9)$$

Another possibility to construct a unitary matrix from the anti-hermitian W is to

use the so-called Cayley transform, which is defined as

$$U = (I - W)^{-1}(I + W) , \quad (3.10)$$

where I is the identity matrix. The inverse transformation is given by

$$W = (U - I)(U + I)^{-1} . \quad (3.11)$$

One should note that we cannot generate any generic element of $SU(n)$ by this transformation, since we are excluding certain matrices, for example, those which are unitary but not anti-hermitian. Both options, the Cayley and exponential map, take $\mathcal{O}(n^3)$ time to compute, which will become inefficient for large matrices.

3.3 Kullback-Leibler divergence

In order to train a flow network for an inference or modeling task, an appropriate choice for the loss function has to be made. The most common choice is the relative entropy or Kullback-Leibler divergence. It serves as a measure for how much two distributions differ from each other. It is important to know that the divergence is not symmetric in its arguments. For flow networks we can distinguish between the forward KL divergence and the reversed one.

The forward version is defined as

$$\begin{aligned} D_{\text{KL}}(q_{\text{target}}(x) \parallel p_{\text{flow}}(x)) &= \sum_x q_{\text{target}}(x) \log \left(\frac{q_{\text{target}}(x)}{p_{\text{flow}}(x)} \right) \\ &= \mathbb{E}_{x \sim q_{\text{target}}} [\log q_{\text{target}} - \log p_{\text{flow}}] \\ &= \mathbb{E}_{x \sim q_{\text{target}}} [\log q_{\text{target}} - \log |\det J_{G^{-1}}| - \log p_{\text{base}}] , \end{aligned} \quad (3.12)$$

where the expectation value over the target distribution is carried out as an averaging over the training set. The KL-divergence is zero when the two distributions are equal. The divergence can be interpreted as the additional entropy if one uses the distribution p_{flow} instead of the target q_{target} to sample x . The expectation over q_{target} will give us a constant term such that the loss function can be defined as

$$\mathcal{L} = \mathbb{E}_{x \sim q_{\text{target}}} [\log |\det J_G| - \log p_{\text{base}}] . \quad (3.13)$$

It is worth noting that in this approach we do not need to calculate $G(z)$ explicitly during training, although it is, of course, required for sampling. In the holographic network, which will be presented in the next chapter, we do not have a training set at hand. The evaluation of the expectation value is therefore not possible. However, we can use the relative entropy function with reversed arguments such that the loss includes an expectation value over the network distribution, i.e. the averaging over a sampled data set.

The reversed KL divergence is more suitable for our purposes and is defined as

$$\begin{aligned} D_{\text{KL}}(p_{\text{flow}}(x) \parallel q_{\text{target}}(x)) &= \mathbb{E}_{x \sim p_{\text{flow}}} [\log p_{\text{flow}} - \log q_{\text{target}}] \\ &= \mathbb{E}_z [\log p_{\text{base}} - \log |\det J_G| - \log q_{\text{target}}] , \end{aligned} \quad (3.14)$$

where we abbreviated the notation for the expectation value as $\mathbb{E}_z \equiv \mathbb{E}_{z \sim p_{\text{base}}}$. For the evaluation of the expectation value of q_{target} , a function proportional to the target distribution $\tilde{q}_{\text{target}} \sim q_{\text{target}}$ is sufficient because the normalization constant will just give a constant term due to the logarithm. Our loss function can then be chosen to be

$$\mathcal{L} = \mathbb{E}_z [\log p_{\text{base}} - \log |\det J_G| - \log \tilde{p}_{\text{target}}] . \quad (3.15)$$

Training of the flow under the reversed KL-divergence means that we are trying to find an approximation of q_{target} by fitting a function p_{flow} to the target distribution. This process is called probability density distillation.

There exist an interesting connection between the training procedures under the two versions of the KL-divergence. Let us first define the probability distribution $q_{\text{norm}}(z)$, which is the density induced by the network transformation via normalizing $q_{\text{target}} \xrightarrow{G^{-1}} q_{\text{norm}}$. It can be found by changing variables from x to z for $q_{\text{target}}(x)$:

$$q_{\text{norm}}(z) = q_{\text{target}}(x) |\det J_G|^2 . \quad (3.16)$$

A forward KL-divergence can be obtained from the reversed version by using the above transformation:

$$\begin{aligned} D_{\text{KL}}(p_{\text{flow}} \parallel q_{\text{target}}) &= \mathbb{E}_z [\log p_{\text{base}}(z) - 2 \log |\det J_G(z)| - \log q_{\text{target}}(G(z))] \\ &= \mathbb{E}_z [\log p_{\text{base}}(z) - \log q_{\text{norm}}(z)] \\ &= D_{\text{KL}}(p_{\text{base}}(z) \parallel q_{\text{norm}}(z)) . \end{aligned} \quad (3.17)$$

We see that fitting the distribution $p_{\text{flow}}(x)$ to the target $q_{\text{target}}(x)$ under the reversed KL-divergence is equivalent to fitting the induced density $q_{\text{norm}}(z)$ to the base distribution under the forward divergence.

3.4 Reparametrization trick

The typical choice for the base density is a normal distribution. In the simplest case, each variable $z^\alpha \in \mathbb{C}$ is drawn from $\mathcal{N}_{\mathbb{C}}(0, 1)$ with density

$$p(z^\alpha) = \frac{1}{\pi} e^{-|z^\alpha|^2} . \quad (3.18)$$

Effectively, the real and imaginary part of z^α are sampled from real normal distributions with variance $\frac{1}{2}$, i.e. $\text{Re}(z^\alpha), \text{Im}(z^\alpha) \sim \mathcal{N}(0, \frac{1}{2})$. All variables z^α are i.i.d. and have no correlation among them, so

$$\mathcal{C}(z^\alpha, z^\beta) = \mathbb{E}[z^\alpha, z^\beta] = 0 \quad \forall z^\alpha, z^\beta . \quad (3.19)$$

This puts a lot of restriction on the information, that can be stored in the base distribution. A less biased choice is a multivariate complex Gaussian

$$p_{\text{base}}(z) = \frac{1}{\det(\pi\Sigma)} \exp\left[-\frac{1}{2} z^\dagger \Sigma^{-1} z\right] , \quad (3.20)$$

where $\Sigma = \mathbb{E}[zz^\dagger]$ is the covariance matrix. Its off-diagonal components contain the correlation between different random variables z^α . Because this distribution has zero mean ($\mathbb{E}[z] = 0$) and zero pseudo-covariance ($\mathbb{E}[zz^T] = 0$), it has circularly symmetry, i.e. z is evenly distributed across all directions in the complex plane.

The matrix Σ can be made learnable via the reparametrization trick. Since the evaluation of the loss function involves the expectation value over the base distribution, it is difficult to train parameters contained in p_{base} . The idea is to implement the multivariate Gaussian via the use of a linear map. We observe that one can obtain $z \sim p_{\text{base}}(z)$ from a transformation of i.i.d. variables $\epsilon \sim \mathcal{N}_{\mathbb{C}}(0, 1)$ via $z = L\epsilon$. The lower triangular matrix L can be calculated from the Cholesky decomposition of the covariance and satisfies $\Sigma = LL^\dagger$. We can make a quick check by changing variable

for the base distribution to get

$$p_{\text{base}}(\epsilon) = \frac{1}{\pi^N} \exp\left[-\frac{1}{2}\epsilon^\dagger\epsilon\right], \quad (3.21)$$

where the covariance matrix has become the identity and all correlations between different random variables have been removed.

4 Holographic RG with Flow Networks

When modeling a boundary theory and its holographic bulk with a neural network, the first step is to decide which physical quantity will be represented by the network nodes. One possibility is to use the Hamilton formulation of the gravity dual and identify the units of a sequential net with the values of the bulk field and its conjugate momentum [15]. The forward direction of the network will then represent the radial coordinate z of the bulk space and the weight matrices will be constrained to produce discrete Hamilton equations of motion. To train the model, a set of observables in the boundary QFT has to be mapped to the first layer of the network because the inputs are boundary values of the bulk fields. This has been demonstrated for a holographic QCD theory [16]. The disadvantage of this approach is that the neural network is defined on the phase space of the fields and will therefore not directly model a holographic spacetime, i.e. it is not a discretization of AdS space. The geometric information is contained in the derivatives appearing in the equations of motion, which are in turn encoded as the weight constraints. The phase space representation also restricts the size of the sequential net, i.e. the input dimension is fixed to two in the case of one scalar bulk field. Moreover, sampling will be quite difficult or inefficient, as we have no access to the probability distribution of the CFT.

Another option comes from exploiting the fact, that the visible units of a trained restricted Boltzmann machine (RBM) obey the distribution $p \sim e^{-E}$, which can be used to model a QFT, when the energy function is identified with the classical limit of the gravity action [14]. The connections between the layers of the RBM have to be carefully defined by choosing a consistent discretization scheme for the field derivatives, that appear in the action. Further constraints on the network units are also needed to ensure the existence of a saddle approximation for the gravity theory.

The RBM can then learn information about the bulk space geometry because some metric components appear in the weights. The network size can be easily scaled because it is defined in discrete real space.

While this generative model gives us the ability of sampling, the probability distribution is unfortunately not tractable. To circumvent this problem we will use a normalizing flow instead, which is much more efficient and versatile because of its invertibility and tractable probability density. But unlike before, we will not model the bulk fields and then infer CFT observables from the boundary values, but we will try to map the CFT degrees of freedom directly to a bulk field. To achieve that, the network will be set up as a renormalization procedure for the boundary QFT. Since we will learn the probability distribution via the use of the reversed KL divergence, there is no need to prepare a training data set. The hyperbolic geometry will be an emergent feature, that can appear after successful training.

4.1 General architecture

We design the holographic flow with a structure known from tensor networks, the multi-scale entanglement renormalization ansatz (MERA). It was proposed as a method to find the ground state of a critical quantum system [27]. When a wave function is fed to the network, degrees of freedom of a certain length scale will be separated at each layer, where the layer depth corresponds to said scale. Therefore, MERA performs a renormalization on the input data and each layer is an RG step. Such a neural network was first implemented in [21], where it was applied to spin states. It was later generalized to other data sets like CelebA, CIFAR-10, or MNIST [19], which are more common in the machine learning community. The basis of our discussion and analysis will be [18], where the neural RG was applied to a holographic setup. Our network will be similar to those implementations, namely it will have the same MERA architecture, but has some modifications like different choices of transformations.

The flow has the MERA structure shown in figure 4.1. The depth of the network can be interpreted as the holographic direction. The deeper we go into the interior, the lower the energy scale $\Lambda = z^{-1}$ of the renormalized CFT becomes. Similar to CNNs the input data will be processed in patches of a specific kernel size $\sqrt{n} \times \sqrt{n}$. On these n variables, a set of transformations is applied consisting of a scaling map,

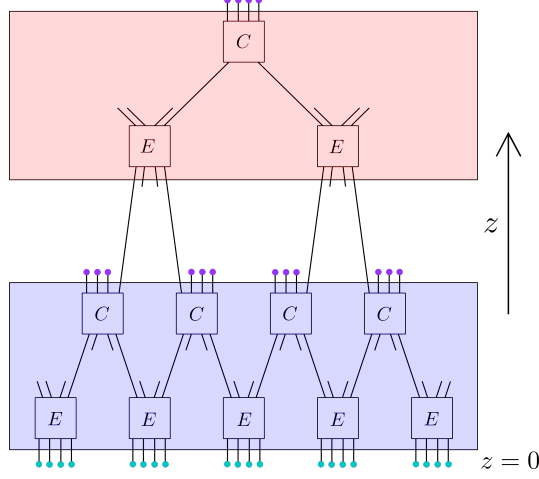


Figure 4.1: General architecture of the holographic flow. The colored boxes are RG-layers at different energy scales. The violet dots are bulk variables and the teal ones are CFT variables.

linear layer, and an activation function. There are two types of transformations: the disentanglers E and the decimators C . The map E serves to reduce the correlation between variables, while C performs a coarse-graining in real space. After a decimator transformation only $\frac{n}{4}$ of the n incoming variables will be processed in the next layer. The other three fourth of the degrees of freedom are taken to be the bulk variables sampled from the base distribution. As the standard kernel size we will choose 2×2 and work with this in the following. An RG-step consists of two layers: one with only disentanglers and the other with only decimators. For two-dimensional data we can distribute E and C in an overlapping manner, see figure 4.1.

We will use a physics motivated notation for the discussion of the network: boundary variables are denoted by the CFT field ψ , the latent variables will be referred to as the bulk field ϕ and the base density will be called the bulk distribution p_{bulk} . The training will happen in two stages. In stage I we will sample the bulk variables from i.i.d. normal distributions and train the network using the generative direction G . This is the standard approach in probability density distillation with normalizing flows. After training, the network weights in G will be fixed and we will relax the bulk distribution to be a multivariate Gaussian. We will then try to capture the residual correlation of the bulk variables by training the covariance matrix via the reparametrization trick. The off-diagonal components of Σ will then contain the information about the hyperbolic geometry.

Exact holographic mapping An analytic version of the network above was given in [24] and [20]. It shares a similar structure as MERA, but with the difference that the RG-steps are only composed of one layer instead of two. In particular, the disentanglers have been omitted. It was shown that by choosing the linear map to be a unitary matrix one can coarse-grain a boundary Hamiltonian. The map actually performs a basis change in Hilbert space on the operators. Assuming n incoming operators, the unitary map separates the degrees of freedom into UV and IR modes. When going from the boundary into the bulk interior, the IR degrees of freedom are fed into the next layer of transformation, while the UV modes are interpreted as bulk operators. This corresponds to a UV-cutoff of the CFT. An explicit matrix can be given for which the Hamiltonian will split into three terms. One of them is simply a rescaling of the original Hamiltonian and can be identified as the IR mode, since it is the result of a renormalization. The other two terms are then the UV mode and an interaction term. Consider a system on a lattice with an action of the form

$$S = \sum_x \psi_x^* \hat{V}_x \psi_x , \quad (4.1)$$

where $\hat{V}_x$ is some interaction operator. A coarse-graining can be easily achieved by grouping together two neighboring fields at sites $(2x)$ and $(2x-1)$ and defining the new degrees of freedom to be proportional to the sum. A rotation like

$$\begin{pmatrix} \psi_x^{IR} \\ \psi_x^{UV} \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} \psi_{2x-1} \\ \psi_{2x} \end{pmatrix} \quad (4.2)$$

can map ψ_x into renormalized modes ψ_x^{IR} and bulk modes ψ^{UV} . If we rewrite the action with ψ_x^{IR} and ψ^{UV} , the term, which depends only on ψ_x^{IR} , will be of the same form as the original action except being rescaled with half the number of operators:

$$S_{\text{renorm}} = \frac{1}{2} \sum_x \psi_x^{*IR} \hat{V}_x \psi_x^{IR} . \quad (4.3)$$

Thus, the boundary theory has been renormalized. We can then perform the transformation (4.2) iteratively at each layer of the network on the IR modes ψ_x^{IR} coming from the previous layer. With the precise mapping of boundary degrees of freedom to bulk operators one has obtained an exact holographic mapping (EHM).

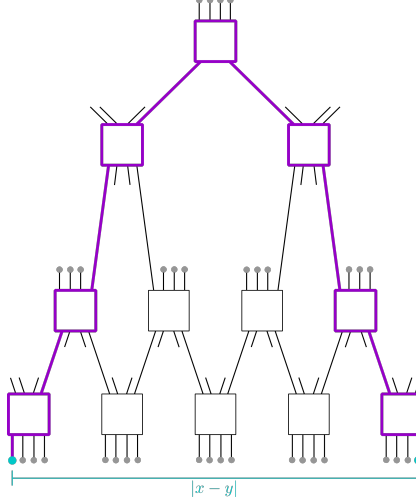


Figure 4.2: The violet geodesic is the path with the least number of links connecting the two teal CFT sites. The distance $|x - y|$ on the boundary is the separation in terms of site numbers.

Hyperbolic geometry The MERA architecture actually exhibits a geometric structure, which is suitable for holography. Let us look at the geodesic distance between two sites on the boundary, i.e. the shortest path connecting them in the bulk (see figure 4.2). We observe that the geodesic has to decay logarithmically with the boundary distance as

$$\mathcal{D}_{\text{MERA}}(x, y) = \alpha \log |x - y| \quad (4.4)$$

with some proportionality factor α . This is consistent with a hyperbolic space. To check the ability of the network to model a CFT, we analyze the two-point function. Let's denote a site on the boundary at position x by ψ_x . The correlation between two sites at x and y in general decays exponentially in terms of the geodesic distance [11]

$$\langle \psi_x \psi_y^* \rangle \sim e^{-m \mathcal{D}_{\text{MERA}}(x, y)} , \quad (4.5)$$

into which we can plug in the expression from before. This leads to a polynomial decay of the correlation as one would expect of a system in critical phase:

$$\langle \psi_x \psi_y^* \rangle \sim e^{-m \alpha \log |x - y|} \sim \frac{1}{|x - y|^{-m \alpha}} . \quad (4.6)$$

It seems that MERA is a good candidate for a discrete bulk space in a holographic setup. In order for this AdS/MERA correspondence to be consistent, one should identify the lattice spacing with the AdS radius L by matching geodesic lengths [5].

This means that the network can only capture physics on scales larger than L and should be regarded as a classical spacetime.

4.2 Discrete field theories

Let us elaborate on the physical theory, which we want to model with the network. We will work with bosonic fields on a square lattice with N sites in total. To avoid irregular behavior at the edges, we impose periodic boundary conditions. We use Poincaré coordinates, where the boundary coordinates are denoted by $x, y \in \mathbb{R}^2$ and the radial direction by $z \in \mathbb{R}$.

4.2.1 Boundary CFT

We choose a complex scalar field ψ living two-dimensional real space whose dynamics is governed by the action

$$S_{\text{CFT}}[\psi] = -k \sum_{\langle x, y \rangle} \psi_x^* \psi_y + \sum_x (\mu |\psi_x|^2 + \lambda |\psi_x|^4) , \quad (4.7)$$

where the sum $\sum_{\langle x, y \rangle}$ should to be taken over nearest neighbors and $(k, \mu, \lambda) \in \mathbb{R}$ are global parameters. The associated probability distribution is given by

$$q_{\text{CFT}}(\psi) = \frac{1}{Z_{\text{CFT}}} e^{-S_{\text{CFT}}[\psi]} . \quad (4.8)$$

If one wants to make connection to Witten's description of holography, one can add a further term which couples the field ψ to the boundary value of a bulk field $\phi_0 \equiv \phi(z=0)$. Such a source term with minimal coupling has the form

$$S_{\text{src}}[\psi] = -\frac{1}{2} \sum_x (\psi_x^* \phi_{0x} + \phi_{0x}^* \psi_x) . \quad (4.9)$$

In the following we will work with the case of vanishing external source $\phi_0 \rightarrow 0$. For the parameters we choose $k = 5$, $\mu = -180$, $\lambda = 25$, which creates a mexican hat potential for ψ with $U(1)$ symmetry and a reflection symmetry $\mathbb{Z}_2$ (see figure 4.3). During the training process, the field $\psi = \rho e^{i\theta}$ will condensate into the minima and is then constrained to have a approximate constant amplitude $|\psi| \approx 2$, such that we can ignore the fluctuations perpendicular to the θ direction. Then only one degree

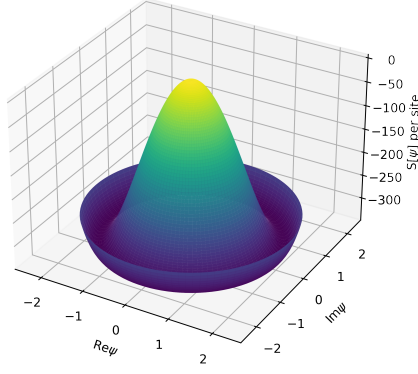


Figure 4.3: Mexican hat potential in the complex plane
with minima at $\rho = \sqrt{\frac{37}{10}}$ and $S_{min} = -\frac{1369}{4}$

of freedom θ survives and the boundary theory effectively becomes an XY-model with action

$$S_{\text{CFT}} \approx -k\rho^2 \sum_{\langle x,y \rangle} \cos(\theta_x - \theta_y) \quad (4.10)$$

at the temperature $T = \frac{1}{k\rho^2}$.

Low temperature limit It is important to mention that this theory is in general not conformally invariant. While the Mermin-Wagner theorem prohibits any spontaneous symmetry breaking in $d \leq 2$, the system can still experience a topological phase transition. This means that there exists a critical phase at low temperatures, where the system exhibits a quasi-long-range interaction and can thus be described by a CFT. There is a certain temperature T_{KT} , below which the system can form bound pairs of vortices and goes into an ordered phase. The correlation function will obey a power law instead of an exponential decay, which implies an infinite correlation length. This indicates the existence of some conformal symmetry. To check this, we will look for an expression of the two-point correlator in the critical phase. Since for low T the fluctuations of θ are small, we can expand the action (4.10) up to quadratic order as

$$-k\rho^2 \sum_{\langle x,y \rangle} \cos(\theta_x - \theta_y) = -2Nk\rho^2 + \frac{1}{2}k\rho^2 \sum_{\langle x,y \rangle} (\theta_x - \theta_y)^2 + \dots \quad (4.11)$$

Within this harmonic approximation we expect small variations of the fields, i.e. $\theta(x) \ll 1$, such that the subtraction can be seen as a derivative by taking the

continuum limit. So we arrive at a free field theory for $\theta(x)$ with

$$S_{\text{CFT}} \approx \frac{k\rho^2}{2} \int d^2x (\nabla\theta(x))^2. \quad (4.12)$$

The two-point function can be found from the Green's function of the Laplace equation in d dimensions and reads

$$\begin{aligned} \langle \theta(x)\theta(y) \rangle &= -\frac{1}{k\rho^2} \frac{|x-y|^{2-d}}{(2-d)S_d} + \text{const.} \\ &\xrightarrow{d=2} -\frac{1}{2\pi k\rho^2} \ln|x-y| + \text{const.} \end{aligned} \quad (4.13)$$

with S_d being the solid angle. The two-point correlation of our original field ψ can then approximated as

$$\begin{aligned} \langle \psi_x \psi_y^* \rangle &\sim \langle e^{i(\theta_x - \theta_y)} \rangle \\ &= e^{-\frac{1}{2}\langle (\theta_x - \theta_y)^2 \rangle} \\ &\sim \frac{1}{|x-y|^{\frac{1}{2\pi k\rho^2}}}, \end{aligned} \quad (4.14)$$

where the second line is valid, since we have a Gaussian distribution due to (4.12). The same estimation can be done for the real and imaginary part and for those we get

$$\begin{aligned} \text{Im} \langle \psi_x \psi_y^* \rangle &\approx 0 \\ \text{Re} \langle \psi_x \psi_y^* \rangle &\sim \left| \langle \psi_x \psi_y^* \rangle \right| \sim \frac{1}{|x-y|^{\frac{1}{2\pi k\rho^2}}}. \end{aligned} \quad (4.15)$$

This power-law is consistent with the result from a CFT. For the XY-model it means that there is an ordered phase at low T and the system can undergo a phase transition, the so-called Kosterlitz-Thousless transistion. Our parameter choice leads to a temperature of $T \approx 0.05$, which is much lower than the estimated critical temperature at $T_{\text{KT}} \approx 0.9$. The emergence of a dual gravity theory in the network can therefore be expected.

4.2.2 Bulk gravity theory

The physical system on the gravity side can potentially contain a large set of different fields. We assume that on large scales the general behavior can be described

effectively by a massive complex scalar field $\phi(x, z)$ with action

$$S_{\text{bulk}}[\phi] = \int d^d x dz \sqrt{g} \left[g^{\mu\nu} \partial_\mu \phi^* \partial_\nu \phi + m^2 |\phi|^2 \right] , \quad (4.16)$$

where higher order interactions are suppressed. Since this action is quadratic in ϕ , we can discretize it with a general ansatz

$$S_{\text{bulk}}[\phi] \approx \sum_{\alpha, \beta} \phi_\alpha^\dagger \Sigma_{\alpha\beta}^{-1} \phi_\beta = \phi^\dagger \Sigma^{-1} \phi , \quad (4.17)$$

where greek indices go over the bulk coordinates, i.e. $\alpha = (x, z)$. This leads to a circularly-symmetric complex normal distribution of N random variables

$$p_{\text{bulk}}(\phi) = \frac{1}{\det(\pi \Sigma)} e^{-\phi^\dagger \Sigma^{-1} \phi} . \quad (4.18)$$

Because the covariance matrix Σ contains the geometric information about the bulk space, we have to make it trainable by the reparametrization trick. The base distribution will then be chosen to be a complex normal distribution with $\Sigma = 1$, i.e.

$$\tilde{p}_{\text{bulk}}(\varphi) = \frac{1}{\pi^N} e^{-\varphi^\dagger \varphi} , \quad (4.19)$$

and the bulk field can be obtained by

$$\phi = L \varphi , \quad (4.20)$$

where L can be calculated from the Cholesky decomposition of the covariance matrix $\Sigma = LL^\dagger$.

4.2.3 Geodesics

Since we know that geodesics in AdS space between two points (x, y) at the same radial position z_0 go as $\mathcal{D}_{\text{geo}} \sim \ln|x - y|$, we can check for a hyperbolic geometry by analyzing the functional dependence of geodesics in our network. The distance in the bulk space can be inferred from the correlator on large scales. If the bulk field is massive, the two-point function will decay exponentially with the distance between two points in the bulk as

$$\langle \phi_\alpha \phi_\beta^* \rangle \sim \exp \left[-\frac{\mathcal{D}_{\text{holo}}(\phi_\alpha, \phi_\beta)}{\xi} \right] , \quad (4.21)$$

where ξ is the correlation length. This can be used as a definition of the geodesic distance because the correlations can be extracted from the covariance matrix $\Sigma_{\alpha\beta} = \langle \phi_\alpha \phi_\beta \rangle$ or from a batch of sampled bulk states as $\langle \phi_\alpha \phi_\beta^* \rangle = \mathbb{E}_\phi[\phi_\alpha \phi_\beta^*]$. So we define

$$\mathcal{D}_{\text{geo}}(\phi_\alpha, \phi_\beta^*) = -\xi \ln \frac{|\langle \phi_\alpha \phi_\beta^* \rangle|}{C_0} , \quad (4.22)$$

where C_0 is a normalization constant which we have to determine. The offset C_0 is chosen as the average value of the diagonal entries of the covariance. We expect that Σ will have roughly the same value for every variable and that the diagonal values are much larger than the off-diagonals. This would be the case of a very massive field, which was our assumption for (4.21). The constant can then be estimated to be

$$C_0 = \frac{1}{N} \sum_\alpha |\langle \phi_\alpha \phi_\alpha^* \rangle| \approx |\langle \phi_\alpha \phi_\alpha^* \rangle| . \quad (4.23)$$

This choice will set the holographic distance between bulk variables at the same position to approximately zero:

$$\mathcal{D}_{\text{geo}}(\phi_\alpha, \phi_\alpha) = -\xi \ln \frac{|\langle \phi_\alpha \phi_\alpha^* \rangle|}{C_0} \approx 0 \quad (4.24)$$

for any ϕ_α .

Mutual information The definition of distance as (4.22) has the disadvantage that it depends on the particular pair of field variables and we might overlook correlations. A maybe more reliable option is the mutual information defined as

$$I(\phi_\alpha : \phi_\beta) \equiv H(\phi_\alpha) + H(\phi_\beta) - H(\phi_\alpha, \phi_\beta) , \quad (4.25)$$

where H is the entropy. We can use I to bound the correlator from above. To do so, we notice first that the mutual information can also be expressed by the Kulback-Leibler divergence as

$$I(\phi_\alpha : \phi_\beta) = D_{\text{KL}}(p(\phi_\alpha, \phi_\beta) \parallel p(\phi_\alpha)p(\phi_\beta)) . \quad (4.26)$$

By using Pinsker's inequality for the KL-divergence we can get a bound for the two-point function:

$$\begin{aligned}
I(\phi_\alpha : \phi_\beta) &\geq \frac{1}{2} \left(\sum_{\phi_\alpha, \phi_\beta} |p(\phi_\alpha, \phi_\beta) - p(\phi_\alpha)p(\phi_\beta)| \right)^2 \\
&= \frac{1}{2} \left(\sum_{\phi_\alpha, \phi_\beta} |p(\phi_\alpha, \phi_\beta) - p(\phi_\alpha)p(\phi_\beta)| \left| \frac{\phi_\alpha \phi_\beta^*}{\|\phi_\alpha\|_\infty \|\phi_\beta^*\|_\infty} \right| \right)^2 \\
&= \frac{1}{2} \left| \sum_{\phi_\alpha, \phi_\beta} [p(\phi_\alpha, \phi_\beta) \phi_\alpha \phi_\beta^* - p(\phi_\alpha) \phi_\alpha p(\phi_\beta) \phi_\beta^*] \frac{1}{\|\phi_\alpha\|_\infty \|\phi_\beta^*\|_\infty} \right|^2 \quad (4.27) \\
&\geq \frac{1}{2} \left| \sum_{\phi_\alpha, \phi_\beta} [p(\phi_\alpha, \phi_\beta) \phi_\alpha \phi_\beta^* - p(\phi_\alpha) \phi_\alpha p(\phi_\beta) \phi_\beta^*] \frac{1}{\|\phi_\alpha\|_\infty \|\phi_\beta^*\|_\infty} \right|^2 \\
&= \frac{1}{2} \frac{|\langle \phi_\alpha \phi_\beta^* \rangle|^2}{\|\phi_\alpha\|_\infty^2 \|\phi_\beta^*\|_\infty^2} ,
\end{aligned}$$

where we assumed $\langle \phi_\alpha \rangle = \langle \phi_\beta \rangle = 0$ in the last line.

Let us calculate the mutual information for a bivariate complex Gaussian to get a formula for the information between two variables. The bivariate distribution is the effective density for two sites and can be obtained by marginalization. The joint entropy for a complex multivariate normal distribution of N variables with covariance Σ can be calculated as

$$\begin{aligned}
H(\phi) &= - \int d\phi d\phi^* p \ln p \\
&= \ln |\det(\pi\Sigma)| + \langle \phi^\dagger \Sigma^{-1} \phi \rangle \\
&= \ln |\det(\pi\Sigma)| + \langle \text{tr}(\phi \phi^\dagger \Sigma^{-1}) \rangle , \quad (4.28) \\
&= \ln |\det(\pi\Sigma)| + \text{tr} \langle \phi \phi^\dagger \Sigma^{-1} \rangle \\
&= N \ln(\pi) + \ln |\det(\Sigma)| + N .
\end{aligned}$$

where we used the definition $\Sigma = \langle \phi \phi^\dagger \rangle$ to evaluate the trace in the third line. Plugging this expression into equation (4.25) gives us

$$I(\phi_\alpha : \phi_\beta) = \ln \left(1 - \frac{\langle \phi_\alpha \phi_\beta^* \rangle \langle \phi_\beta \phi_\alpha^* \rangle}{\langle \phi_\alpha \phi_\alpha^* \rangle \langle \phi_\beta \phi_\beta^* \rangle} \right) , \quad (4.29)$$

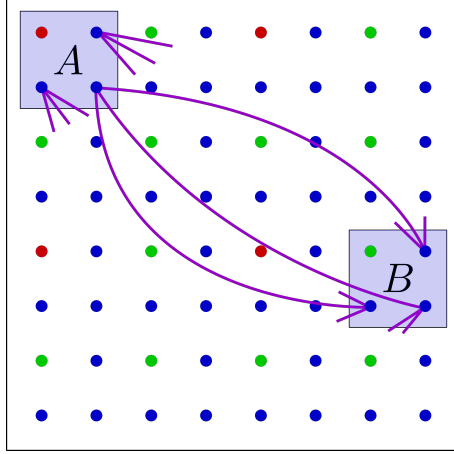


Figure 4.4: Distribution of bulk variables. Red dots represents IR modes, blue ones are UV modes and green dots are of intermediate length scale. The distance between two UV filters A, B is the average of the 9 paths connecting the associated blue dots.

which presents an alternative way to calculate the geodesic distance

$$\mathcal{D}_{\text{geo}}(\phi_\alpha, \phi_\beta) = -\xi \ln \frac{I(\phi_\alpha : \phi_\beta)}{I_0} , \quad (4.30)$$

where I_0 is a normalization constant corresponding to the case of maximal entanglement.

While the bulk sample will have the same shape as the boundary data on the lattice, the bulk variables are actually spread in the radial direction z , i.e. they belong to different length scales of the RG-flow. The specific assignment to the energy scales is dependent on the kernel size of the decimator map C . A hypersurface at fixed z will therefore not be evenly covered by the bulk variables, see figure 4.4. In order to define a holographic distance in a consistent way, we define the holographic distance to be evaluated between different patches covered by a decimator filter. The distance between two filters A, B will then be the average geodesic distance between the bulk variables associated to it and can be defined as

$$\mathcal{D}_{\text{holo}}(A, B) = \text{avg}_{\phi_\alpha \in A, \phi_\beta \in B} \mathcal{D}_{\text{geo}}(\phi_\alpha, \phi_\beta) . \quad (4.31)$$

To probe the geometry, we will calculate the angular distance within one layer and the radial distance across different z . The radial geodesic has to be proportional to

the number of layers r separating two decimators like

$$\mathcal{D}_{\text{holo}}^{\text{rad}}(A, B) \sim r . \quad (4.32)$$

For the angular distance we expect for a hyperbolic space a logarithmic dependence on the difference in the coordinates as

$$\mathcal{D}_{\text{holo}}^{\text{ang}}(A, B) \sim \ln |r| , \quad (4.33)$$

with $r = \sqrt{x^2 + y^2}$, where A and B are separated by x and y decimators in the two lattice directions.

4.3 Flow design

4.3.1 Loss function and saddle point

The probability distribution, that the network produces, can be calculated by a change of variables from the bulk distribution as

$$p_{\text{flow}}(\psi) = p_{\text{bulk}}(\phi) \left| \det \frac{\partial G(\phi)}{\partial \phi} \right|^{-1} \left| \det \frac{\partial G(\phi^\dagger)}{\partial \phi^\dagger} \right|^{-1} . \quad (4.34)$$

Since we transform $\text{Im}(\phi)$ and $\text{Re}(\phi)$ by the exact same map, the last two factors will give us the same value such that $\left| \det J_{G(\phi)} \right| \left| \det J_{G(\phi^\dagger)} \right| = \left| \det J_{G(\phi)} \right|^2$. For probability density distillation the reversed KL-divergence

$$D_{\text{KL}}(p_{\text{flow}}(\psi) \parallel q_{\text{CFT}}(\psi)) = \mathbb{E}_\phi [\ln p_{\text{bulk}}(\phi) - 2 \ln |\det J_G| + S_{\text{CFT}}] + \ln Z \quad (4.35)$$

will be used, where the last term is constant and will be dropped in the implementation. The expectation value is the average $\mathbb{E}_\phi \equiv \mathbb{E}_{\phi \sim p_{\text{bulk}}}$ over a sampled set. The loss function will then be chosen as

$$\mathcal{L} = \mathbb{E}_\phi [\ln p_{\text{bulk}}(\phi) - 2 \ln |\det J_G| + S_{\text{CFT}}[G(\phi)]] . \quad (4.36)$$

The QFT action will be evaluated on the boundary configurations $G(\phi)$, which are sampled from the network.

Let us interpret the above loss function in the context of holography. What does it

mean to minimize the KL-divergence? Let's define the distribution for the gravity theory induced by the flow from the CFT density as

$$q_{\text{grav}}(\phi, G) = q_{\text{CFT}}(G(\phi)) |\det J_G|^2 , \quad (4.37)$$

which depends on the bulk field as well as on the parameters of the flow transformation contained in G . We can think of G as the background geometry, on which the field ϕ lives. As seen in the previous chapter, a training under the reversed KL-divergence is equivalent to training under the forward KL-divergence, where we try to match $q_{\text{grav}}(\phi, G)$ to a prior distribution $p_{\text{bulk}}(\phi)$. From equation (4.37) we also find an expression for the gravity action:

$$S_{\text{grav}}[\phi, G] = S_{\text{CFT}}[\psi] - 2 \ln |\det J_G| . \quad (4.38)$$

Using this, the forward KL-divergence corresponding to the loss (4.36) is then given by

$$\begin{aligned} \mathcal{L} &= D_{\text{KL}}(p_{\text{bulk}}(\phi) \parallel q_{\text{grav}}(\phi, G)) - \text{const.} \\ &= \mathbb{E}_{\phi}[-S_{\text{bulk}}[\phi] + S_{\text{grav}}[\phi, G]] . \end{aligned} \quad (4.39)$$

Since the first term is a fixed distribution, this equates to minimizing $\mathbb{E}_{\phi}[S_{\text{grav}}[\phi, G]]$. Or in other words, training under this loss is equivalent to finding the saddle point approximation of the gravity theory. We can make an estimation for the marginal distribution $q_{\text{grav}}(G)$ of the network's geometry encoded in G by

$$\begin{aligned} \frac{1}{Z_{\text{grav}}} e^{-\mathbb{E}_{\phi}[S_{\text{grav}}[\phi, G]]} &\leq \frac{1}{Z_{\text{grav}}} \mathbb{E}_{\phi} \left[e^{-S_{\text{grav}}[\phi, G]} \right] \\ &\leq \frac{1}{Z_{\text{grav}}} \sum_{\phi} e^{-S_{\text{grav}}[\phi, G]} \equiv q_{\text{grav}}(G) , \end{aligned} \quad (4.40)$$

which holds by Jensen's inequality due to the convexity of the exponential. This implies that the marginalized probability of G is maximized under training, i.e. the geometry contained in G is classical. We therefore expect that fluctuations are of small order and that the bulk space has a smooth behavior at the global minimum of the loss function.

4.3.2 Bijective transformations

The complete flow map G in generative direction consists of separate inverse RG-layers $G = \prod_k G_k$, which themselves contain a layer with decimators C_i and one with disentanglers E_i . In general, those can be all chosen to be different, so $G_k = G_k(\{C_i\}_k, \{E_i\}_k)$. Since our physical system is translation invariant, it is reasonable to have the filters within a layer to share weights, such that G_k depends only on two filters C_k, E_k . One could go even further and impose scale invariance on the EHM, such that all RG-layers have the same transformation $G_{k+1} = G_k$.

C and E can in principle be chosen to be differentiable and invertible map as long as the logarithm of the Jacobian determinant (LDJ) and the inverse transformation can be calculated easily (like in RNVP networks). But we have seen that an EHM can already be achieved by a single matrix. So we expect our flow with two maps C, E to be expressive enough even if we choose them to be constructed with simple matrices W . Additionally, we equip the net with a scaling layer s and an activation function a . The filters are then compositions of the form

$$\begin{aligned} C &= a \circ W_C \circ s \\ E &= a \circ W_E \circ s , \end{aligned} \tag{4.41}$$

where W will be varied by choosing from a number of options.

There is another map L implemented before the first decimator layer in the direction of G , which is the reparametrization. It transforms a multivariate Gaussian with covariance $\Sigma = 1$ into a normal distribution, where the variables are correlated. Here, we also have freedom to constrain the weights in L .

Reparametrization To make the discretized bulk action $S_{\text{bulk}} = \phi^\dagger \Sigma^{-1} \phi$ reflect locality, we can restrict the form of the inverse covariance matrix by setting all entries to zero if two variables are not nearest neighbors within an RG layer or across different energy scales. The matrix will then have the form introduced in [18]

$$\Sigma^{-1} = \sum_{\langle \alpha, \beta \rangle} \kappa_{\alpha\beta} (|\alpha\rangle\langle\alpha| + |\beta\rangle\langle\beta| - |\alpha\rangle\langle\beta| - |\beta\rangle\langle\alpha|) + mI , \tag{4.42}$$

where I is the identity matrix, m the mass of the field and $\kappa_{\alpha\beta} > 0$. Translation and rotational invariance at constant z can also be implemented if we use the same value $\kappa_{\alpha\beta}$ for all bulk variables within the same layer. This reduces the number of

parameters to only two per RG-step. But this covariance matrix will in general not be positive definite without additional conditions on $\kappa_{\alpha\beta}$ and m . Moreover, it is difficult to constrain Σ in such a way that it stays positive-definite during training. Therefore, we opt for a different approach. Because the Cholesky decomposition into triangular matrices requires the original matrix to be positive-definite, we choose to build the covariance as $\Sigma = LL^\dagger$ and implement a simple linear layer L with the restriction to be lower triangular. The i.i.d. variables $\varphi_\alpha \sim \mathcal{N}_{\mathbb{C}}(0, 1)$ are then transformed to bulk states by

$$\phi = L\varphi . \quad (4.43)$$

This will ensure that Σ will always stay positive definite and ϕ follows a proper probability distribution.

Scaling In the direction of $G(\phi)$ the n incoming variables ϕ_α for one filter will be first scaled by

$$s(\phi_\alpha) = e^{\lambda_\alpha} \phi_\alpha \quad (4.44)$$

with inverse

$$s^{-1}(\psi_x) = e^{-\lambda_x} \psi_x , \quad (4.45)$$

where the LDJ is simply given by the exponent λ .

Linear layer As our reference we choose a general $n \times n$ matrix. Since MERA and the EHM use unitary transformations, we will also implement those. At last we have an orthogonal map, which aims to reproduce the EHM introduced in [24]. The linear layer will then be a matrix from the following groups:

- $GL(n, \mathbb{C})$
- $SU(n)$ from a Cayley transform
- $SU(n)$ from a matrix exponential
- $O(4)$ from stacked $O(2)$ matrices (only for $n = 4$)

It is important to note, that the first three maps have complex weights and the last is restricted to real entries. The $O(4)$ transformation, which was used in [18], can

be build from two block-diagonal matrices

$$Q(\omega_i, \omega_j) = \begin{pmatrix} \cos \omega_i & -\sin \omega_i & 0 & 0 \\ \sin \omega_i & \cos \omega_i & 0 & 0 \\ 0 & 0 & \cos \omega_j & -\sin \omega_j \\ 0 & 0 & \sin \omega_j & \cos \omega_j \end{pmatrix} \quad (4.46)$$

and

$$R(\omega_i, \omega_j) = \begin{pmatrix} \sin \omega_i & \cos \omega_i & 0 & 0 \\ \cos \omega_i & -\sin \omega_i & 0 & 0 \\ 0 & 0 & \sin \omega_j & \cos \omega_j \\ 0 & 0 & \cos \omega_j & -\sin \omega_j \end{pmatrix} \quad (4.47)$$

along with a matrix to permute two variables

$$P = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (4.48)$$

where $R(\frac{\pi}{4}, \frac{\pi}{4})$ reproduces the unitary map from the EHM in its diagonal blocks. The disentangler matrix is then defined as the composition

$$W_E^{O(4)} = Q(\omega_6, \omega_7) \circ P \circ Q(\omega_4, \omega_5) \circ Q(\omega_2, \omega_3) \circ P \circ Q(\omega_0, \omega_1) \quad (4.49)$$

and the decimator as

$$W_C^{O(4)} = R(\omega_6, \omega_7) \circ P \circ R(\omega_4, \omega_5) \circ Q(\omega_2, \omega_3) \circ P \circ Q(\omega_0, \omega_1). \quad (4.50)$$

Activation The non-linearity contains no parameter fro training and is a hyperbolic function

$$a(\phi_\alpha) = \frac{\phi_\alpha}{|\phi_\alpha|} \operatorname{arsinh} |\phi_\alpha|, \quad (4.51)$$

since $\sinh(x)$ naturally appears in AdS spaces. The inverse and LDJ are only properly defined for $\phi_\alpha \neq 0$. In our implementation we choose to set the inverse and LDJ to zero for $\phi = 0$ and get

$$\log \left| \det \frac{\partial a}{\partial \phi_\alpha} \right| = -\frac{1}{2} \ln(1 + |\phi_\alpha|^2). \quad (4.52)$$

In the coarse-graining direction G^{-1} the inverse activation is given by

$$a^{-1}(\psi_x) = \frac{\psi_x}{|\psi_x|} \sinh|\psi_x| \quad (4.53)$$

with

$$\log \left| \det \frac{\partial a^{-1}}{\partial \psi_x} \right| = \ln(\cosh |\psi_x|) . \quad (4.54)$$

4.4 Training and results

The flow was build with the MERA structure from [18] and the ML library PyTorch. We will train various networks with different choices for the linear maps W_C in the decimator and W_E in the disentangler and we will refer to these flows by the combination (W_C, W_E) , e.g. the network (L, L) uses only matrices from $GL(n)$. As the default hyper-parameters we choose:

CFT size	32×32
Kernel size	2×2
Batch size	1024
Stage I steps	60 000
Stage II steps	60 000

The $SU(4)$ matrices are build as parametrizations of a standard `Linear(4,4)` layer from PyTorch. The learning rate for the linear maps is set to 10^{-4} , while for the scalings we decrease the rate by a factor of 10 every 20 000 steps from 10^{-2} to 10^{-4} . The rate for the weight in the reparametrization layer of the Gaussian is set to 10^{-3} . For the models of type (any, L) , we will train additionally with double the number of training steps.

4.4.1 Weight initialization

For the first stage of training we set the weights of the scaling layers to zero to avoid an exploding loss function from the start. Whenever possible, we initialize the linear layers to

$$W_E^{\text{init}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \quad (4.55)$$

and

$$W_C^{\text{init}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 1 & 0 & -1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 1 & 0 & -1 \end{pmatrix}. \quad (4.56)$$

This corresponds to setting the matrices $O(4) \ni Q = 1$ and R to have the EHM transformation on the diagonal blocks, which seems to be a reasonable choice, since this map can produce an RG-scheme. The initialization of the linear layers have to be mapped to the weights of the original `Linear(4,4)` layer by the reverse parametrization. For disentangled with $SU(4)$ matrices from the Cayley map this is not possible, since W_E^{init} is not a solution of the inverse Cayley transform. The $SU(4)$ matrix from the exponential map cannot be specifically initialized at all because there is no matrix logarithm implemented in PyTorch. In these cases we will simply use the default random initial weights. All layers will have no bias term, so they are compatible with the $U(1)$ symmetry of the boundary theory.

For training stage II, the covariance matrix Σ or Cholesky matrix L should not be initiated as an identity matrix. Otherwise, the matrices stay diagonal during training. For our setup we choose to train with the approach of directly using the Cholesky matrix L because the definition of Σ^{-1} with nearest-neighbour interactions does not stay positive-definite during our training. The matrix L will be initialized after training stage I with a small perturbation on the off-diagonal as

$$L^{\text{init}} = \begin{pmatrix} 1 & & 0 \\ & \ddots & \\ \epsilon & & 1 \end{pmatrix}, \quad (4.57)$$

where the entries in ϵ are drawn from a normal distribution $\mathcal{N}(0, 10^{-2})$.

4.4.2 Variation of filters

We train networks with the following combinations of maps:

W_C	L	U_{Cay}	U_{exp}	O	L	L	L	U_{Cay}	U_{exp}	O
W_E	L	L	L	L	U_{Cay}	U_{exp}	O	U_{Cay}	U_{exp}	O

The behavior of the loss function depends on the choice of transformation. The

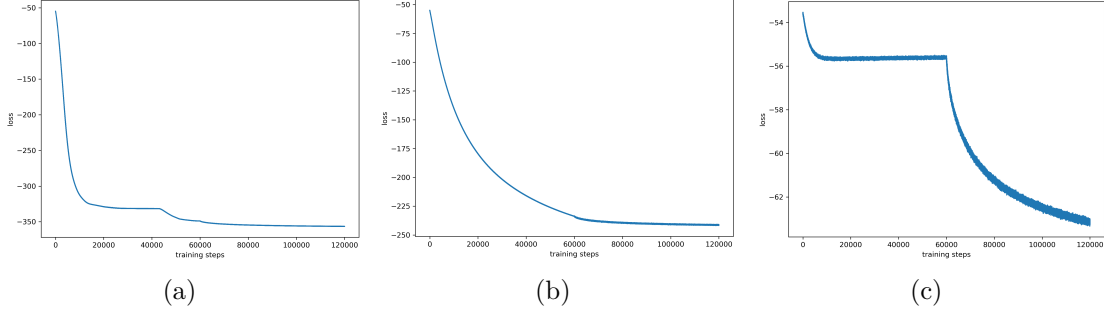


Figure 4.5: Loss for 120k training steps
 (a) *any, L* (b) *L, U/O* (c) *U/O, U/O*

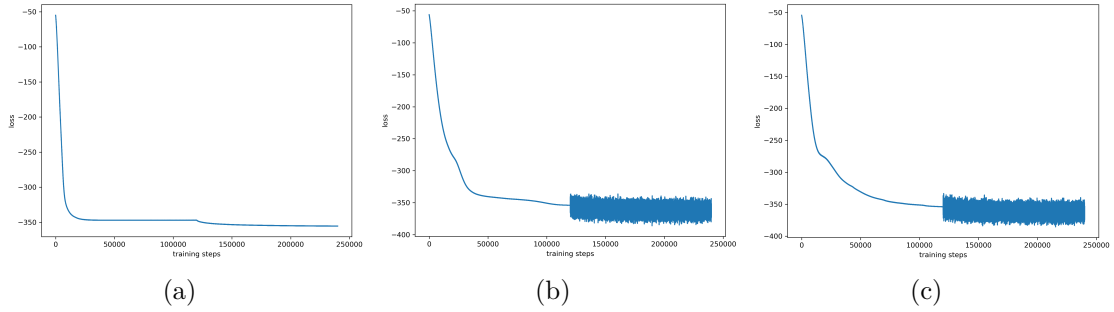


Figure 4.6: Loss for 240k training steps
 (a) *L, L* (b) *U, L* (c) *O, L*

loss is lowest for a $GL(4)$ map in the disentangler and highest when we don't use $GL(4)$ at all, see figure 4.5. In the latter case the model seems to have reached a plateau very quickly, from which we conclude that it is stuck in a local minimum. In all cases the loss is only slowly decreasing after the initial steep drop as usual in neural network training. The flow can be further trained when we enter stage II at 60 000 steps. The loss experience another drop and the covariance matrix of the bulk distribution seems to be capturing some correlation features, which could not be resolved by training the network weights alone. However, this does not necessary imply that these correlations are meaningful. When we increase the training steps to from 120k to 240k, the loss starts fluctuate in stage II for $(U/O, L)$ (see figure 4.6). The geometry apparently cannot be resolved by the Gaussian distribution. The smooth behavior of (L, L) in figure 4.6 will turn out to be some kind of local minimum.

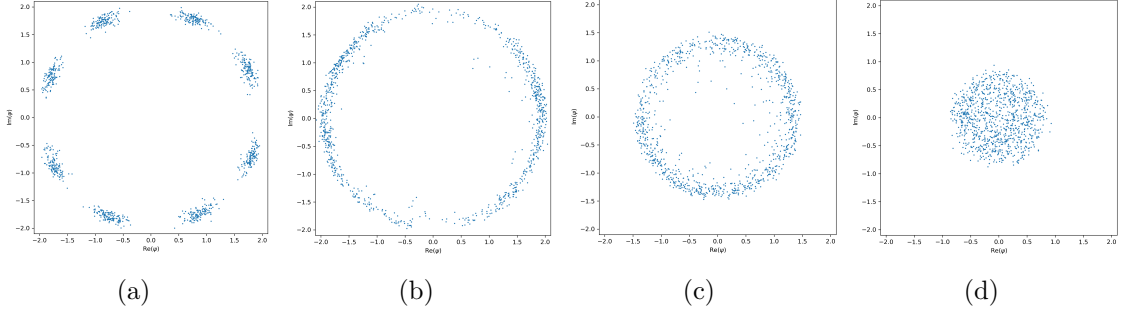


Figure 4.7: Single ψ -distributions for 120k steps
(a) L, L (b) $U/O, L$ (c) $L, U/O$ (d) $U/O, U/O$

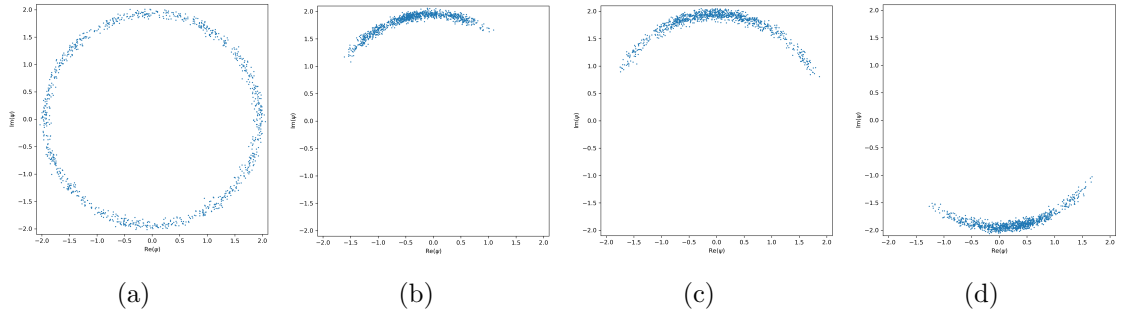


Figure 4.8: Single ψ -distributions for 240k steps
(a) L, L (b) U_{Cay}, L (c) U_{exp}, L (d) O, L

Distributions in the complex plane Let us discuss how these different training behaviors are reflected in the sampled configuration, see figure 4.7. We plot the distributions of a single configuration in the complex plane. Usually, we would expect that a single direction is approximately selected, which will be restored in the ensemble of multiple samples. But the networks (L, L) seem to have formed a periodic setup instead. The distributions for type $(U/O, L)$ hint at a $\mathbb{Z}_2$ reflection symmetry. For the transformations $(any, U/O)$, the field doesn't fully condensate into the circle, i.e. it is not in the lowest energy state. This matches the high value of their loss function. With longer training duration we see that the networks have selected a direction in a single sample, except for L, L . The rotation symmetry is restored if we plot multiple samples at once.

Configurations in real space For the networks, which have reached the minima of the mexican hat potential, we plot the actual configuration in real space (see figure 4.9). The plots for 120k training steps show some alignment but the direction are

still quite evenly distributed. For 240k steps, a long-range alignment can be clearly seen. We see that L, L have collapsed into a mode with periodic structure, which we wouldn't expect from the physical system. This matches their distribution in the complex plane. The (U, L) networks have formed square domains of constant directions and the field in (O, L) has also started to align. When we train these models for 240k steps, we observe that these regions increased in size and the alignments in (O, L) , which seemed to resemble vortices, also became domains. This explains that in the complex plot a certain direction is approximately selected in a configuration.

Two-point correlators The only models with the expected physical properties are $(U/O, L)$ trained with 240k steps. So we will analyze their boundary two-point functions to see whether they are physically sensible. To match the low-temperature behavior of the boundary QFT, the imaginary part should be around zero and $\text{Re}\langle\psi_x\psi_y^*\rangle \approx |\langle\psi_x\psi_y^*\rangle|$, which is the case for these network. The (L, L) type fails to reproduce this result as well as all models trained with 120k steps. To check the dependence of the absolute value of the correlator, we plot $|\langle\psi_x\psi_y^*\rangle|$ in a log-log graph over the distance $|x - y|$. Since we expect a power-law decay, the data points should show a linear dependence. This can almost be observed in all three cases of $(U/O, L)$ except in (U_{Cay}, L) for long distances, see figure 4.10.

Holographic distances For the geodesics in the bulk we use the version defined with the two-point function directly because the one based on mutual information gave frequent errors due to the argument being zero. We plot the radial distances over r and the angular distance within a layer over $\ln|r|$. In both cases a linear relationship should appear, which is unfortunately not the case in any setup. As an example we show the plots for (U_{exp}, L) , see figure 4.11.

Discussion Although some of our networks indeed have shown the correct dependence of the two-point function, this does not necessary imply that the flow has captured all information of the physical dynamics. The theoretical results for $\psi = \rho e^{i\theta}$ are only valid in the low temperature limit, where we have assumed small fluctuations in θ and froze the degree of freedom ρ completely. However, when we look at the distribution in the complex plane of multiple samples for our trained model, we can clearly see that ρ is not completely constrained to be constant but has

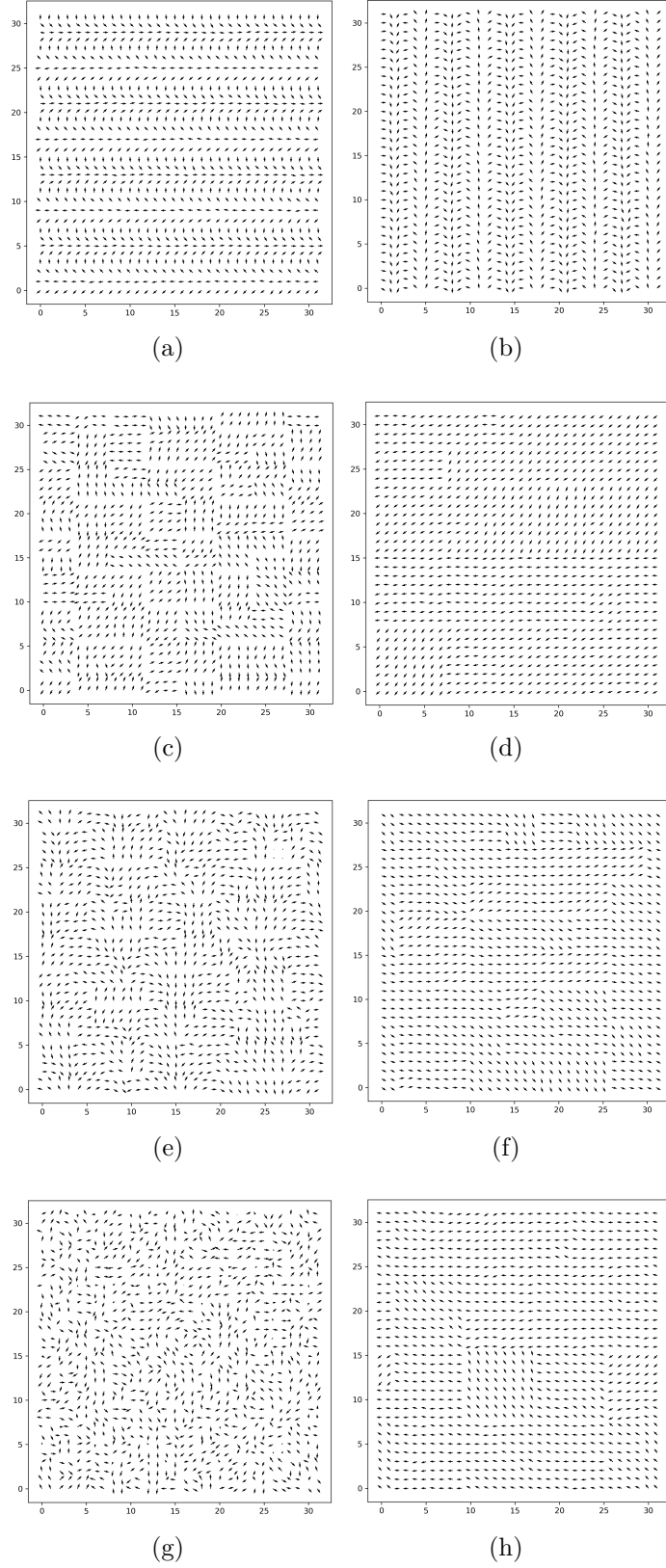


Figure 4.9: Config. for 120k steps: (a) L, L (c) U_{Cay}, L (e) U_{exp}, L (g) O, L
 Config. for 240k steps: (b) L, L (d) U_{Cay}, L (f) U_{exp}, L (h) O, L

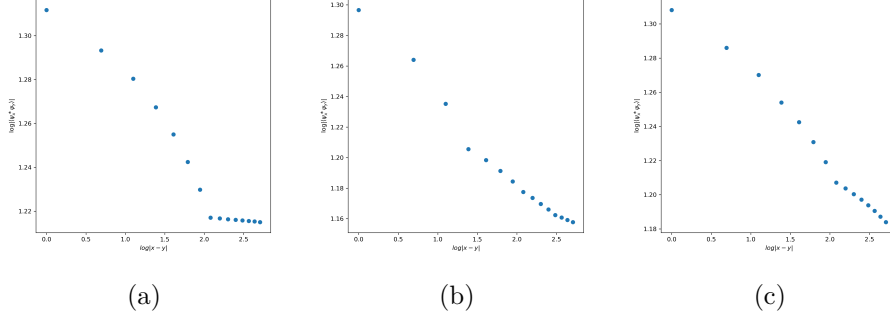


Figure 4.10: $\log|\langle \psi_x \psi_y^* \rangle|$ over $\log|x - y|$
(a) U_{Cay}, L (b) U_{exp}, L (c) O, L

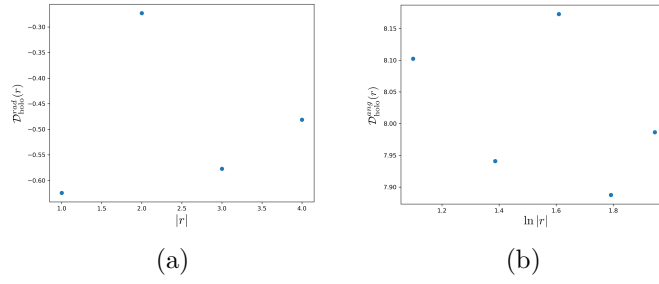


Figure 4.11: Holographic distances for (U_{Cay}, L)
(a) radial (b) angular

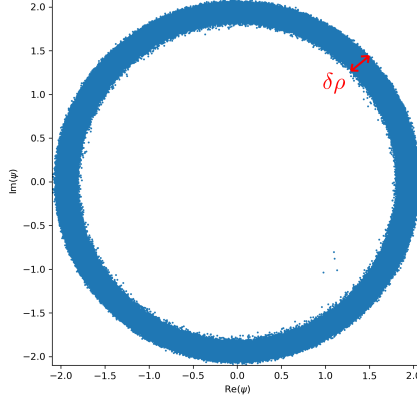


Figure 4.12: Distribution of 1024 samples for (O, L)

fluctuations of order $\delta\rho$, see figure 4.12. So our approximation of the QFT might not be entirely valid for the trained network. This is supported by the fact that we observe the emergence of domain walls in our model instead of vortices, which should appear in the Kosterlitz-Thouless transition. Moreover, a smooth behavior of the bulk geodesics is not necessarily guaranteed, even if we have found a suitable flow configuration matching all phenomena of the XY-model. For our particular CFT one should actually not expect a classical background, which we will explain by a duality argument. Since for the boundary fields we have assumed very small fluctuations in all degrees of freedom, perturbation theory should hold and the CFT is weakly coupled. This means that the dual quantum gravity description should be in the strong coupling regime and therefore far away from its classical limit. This could be reflected in the high fluctuations of the loss function during training stage II, where the covariance tried to capture the information of a classical geometry.

4.5 Model extensions

To further investigate the flow network, we will test two small extensions. The first one is the inclusion of the source term in the action, which has the role of a coupling to an external (magnetic) field. As another modification we implement conformal symmetry into the model.

External field In the training above bias term was set to zero because it would otherwise break the $U(1)$ rotation symmetry of the action. Training with a bias leads

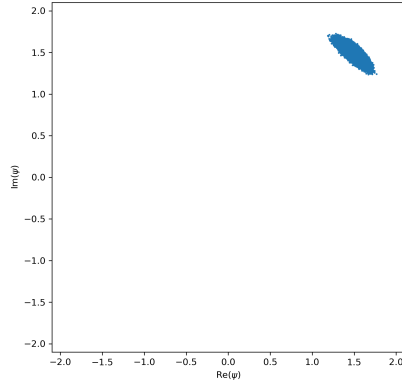


Figure 4.13: 1024 samples with external field $\phi_0 = 100 + 100i$.

to mode collapse, where the system picks a random direction for the fields during training and is then stuck at it. However, if we want to include an external field ϕ_0 , a non-zero bias term is necessary to capture this effect. The network picks up the direction defined by the external source and stays in roughly that configuration over multiple samples, see figure 4.13. Although the ϕ_0 could be seen as a source, it has no connection to the bulk field. It is simply an external parameter and does not correspond to a boundary value of the bulk field ϕ .

Scale invariance Because the boundary CFT is theoretically in critical phase, we could impose scale invariance on the network. As an example, we work with the model (U_{Cay}, L) and use the same filters across all layers. This reduces drastically the number of parameters and results in faster training. But it seems that the flow also struggles to bring the system into the ground state. After training for 120k steps we see that the field has started to condensate into the minima, but much less than the previous networks (see figure 4.14). This could either mean that the additionally constrained flow is not expressive enough or that there is no scale invariance present across the RG-layers. The latter might imply, assuming that the target QFT actually has scale invariance, that the RG-layers do not perform a proper renormalization.

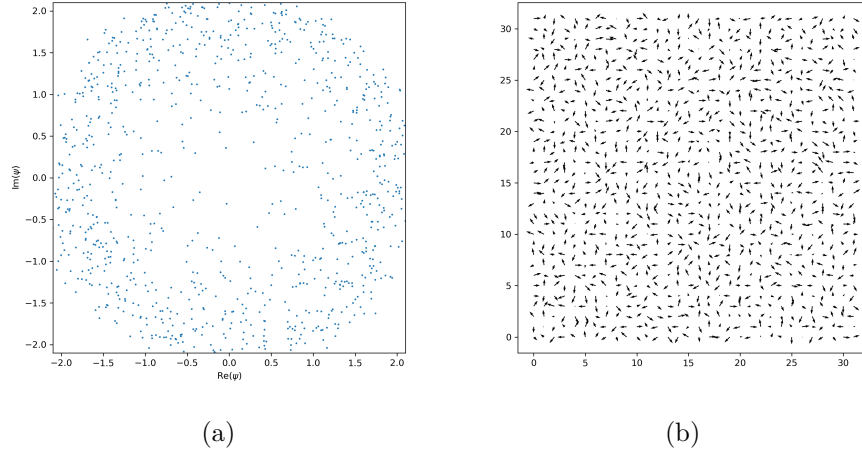


Figure 4.14: (U_{Cay}, L) with scale invariance
(a) distribution in the complex plane (b) field configuration

5 Outlook

In this thesis, we have taken initial steps in constructing and analyzing flow networks with an architecture motivated from the holographic principle of theoretical physics. The probability density of the flow is matched to a target distribution associated with a boundary CFT via probability distillation. In this approach, the neural networks were trained in two stages and the bulk metric is learnable. In the first part, an optimal renormalization scheme for the boundary CFT is found with a base distribution of uncorrelated bulk variables. In the second stage, we encode information about the bulk geometry into the covariance matrix of the bulk density. We achieve this by training the covariance while fixing the weights of the transformations trained in the previous stage. Several maps were implemented in the flow. We have seen that certain restrictions on the weights have to be imposed in order to avoid mode collapse. In particular, using only linear maps was insufficient to ensure a stable training and unitary or orthogonal matrices were needed. The configurations sampled from the trained flow were confirmed to match the target distribution by checking the behavior of the two-point correlators.

Symmetries and equivariance A possible extension in the network design has been demonstrated by including scale invariance. Besides spacetime isometries, one can also add further symmetries to the neural network architecture. We show some possibilities on how to incorporate them. The subfield of machine learning, which covers such implementations, is called geometric deep learning.

The dynamics of physical system is usually restricted by underlying symmetries. In fact, the many fundamental theories in physics can be thought of as a representation theory of their symmetry group. If such a group exists, the system is said to be covariant or equivariant under the group action. When we train a neural network to approximate physical phenomena with a certain symmetry group, the output of the trained network should be equivariant under this group at least approximately.

We can exploit this to boost the training by implementing the isometry into the network architecture. A detailed review of this approach to machine learning is [9]. In the simplest case, we can put a constraint on the network's loss. The network will then tend to respect the symmetry during training. In our model we can modify the probability distribution p_{flow} , which is used in the KL-divergence. Since our holographic theory transforms under reflections $\mathbb{Z}_2$, we can use instead a symmetric probability density

$$\tilde{p}_{\text{flow}}(x) = \frac{1}{2}(p_{\text{flow}}(x) + p_{\text{flow}}(-x)) , \quad (5.1)$$

which was proposed in [21] for an Ising model. While the loss function can now not distinguish between the states x and $-x$ and is therefore equivariant, the network itself and the separate RG-layers are not. This means that the renormalized variables in the flow might not transform equivariantly under $\mathbb{Z}_2$. Implementing symmetries on the level of the layers will not only guarantee the correct behavior of the flow but will also make it easier to interpret the weights and intermediate outputs since they can now represent physical observables. For matrix symmetry groups, a general approach to constrain the weight matrix of a linear layer has been proposed in [12]. Suppose a network layer acts on some input data x by $x' = Wx$. This action is said to be equivariant under a group G if it transforms as

$$W\rho(g)x = \rho'(g)Wx \quad \forall g \in G , \quad (5.2)$$

where ρ, ρ' are some linear representation of g . In a more general case, it means that the matrix W is invariant under the group in some representation $\tilde{\rho}$, i.e.

$$\tilde{\rho}(g)W = W . \quad (5.3)$$

This is equivalent to a constraint using the generators of G

$$d\tilde{\rho}(\mathfrak{g})W = 0 , \quad (5.4)$$

where $\mathfrak{g}$ is an element of the Lie algebra. Using this equation, we can restrict the entries of W such that the layer becomes equivariant. In this way, we could also implement additional symmetries besides $\mathbb{Z}_2$ reflections. For example, if we want to model theories of multiple fields with $\mathcal{N} = 4$ SUSY, we can use this approach to enforce $SU(4)$ equivariance in the RG-layers. The latent outputs in the layers then could be interpreted as physical observables because they are representations of the

symmetry group.

Applications Usually, when training neural networks, it is almost impossible to determine, whether we have reached the lowest value of the loss or are stuck in some local minimum. This is due to the high dimension of the parameter space, on which the loss function is defined. But when the network represents a physical system, we can check for specific properties of the variables like the two-point function in our example. So if all weights and latent variables correspond to physical quantities and they have the correct behavior, we could argue that the loss is close to its global minimum. One might wonder, whether this is a sufficient condition. This is one of the possible use cases for a holographic neural network.

The learned probability distribution p_{flow} of the network can be used to boost the efficiency of Monte Carlo sampling of lattice QCD configurations, which has been proposed in [1] and [2]. To draw samples from q_{CFT} with the usual Metropolis-Hastings algorithm, we use Markov chains, where the sequence can be sampled from an arbitrary density. Suppose we are in a state $\psi^{(i)}$. Then a proposed sample ψ' is accepted with probability $\mathcal{A} = \exp[-\Delta S_{\text{CFT}}]$, where $\Delta S_{\text{CFT}} = S_{\text{CFT}}[\psi'] - S_{\text{CFT}}[\psi^{(i)}]$. In the case of acceptance we set $\psi^{(i+1)} = \psi'$. A special case of the general Metropolis-Hastings uses a proposal probability density to sample ψ' . This proposal density can be taken to be p_{flow} from the trained network. In this MH version, the acceptance ratio is defined as

$$\mathcal{A} = \min \left(1, \frac{p_{\text{flow}}(\psi^{(i)})p_{\text{CFT}}(\psi')}{p_{\text{CFT}}(\psi^{(i-1)})p_{\text{flow}}(\psi')} \right) . \quad (5.5)$$

We can see that almost all samples will be accepted with $\mathcal{A} \rightarrow 1$ if $p_{\text{flow}} \approx p_{\text{CFT}}$. The convergence and efficiency of the Monte Carlo sampling is highly improved, which was demonstrated [21]. The Metropolis-Hastings algorithm serves as correction for our flow sampling, because there is still some discrepancy between p_{flow} and p_{CFT} . In the ideal case of a perfectly trained network, one could sample from the flow directly.

Future steps Although our network could not reproduce a smooth hyperbolic geometry, a correct two-point function behavior of the boundary theory was found. Sampling tasks might be possible with the trained network, but one has to look at other properties of the CFT configurations like higher n-point functions. But the model can also be further analyzed on the bulk side. Apart from looking at

the behavior of geodesics, we could check if the entropy law and Ryu-Takayanagi formula is valid. To do that, we have to find a practical way to calculate the entropy entanglement from the network variables. Such a direction has been taken in [29] based on random tensor networks. Because the geometry of the gravity dual for our choice of field theory might be non-classical, it was difficult to check the validity of our approach. In the future, it would be exciting to test the flow setup for CFTs of which we know the existence of a dual theory with classical background. This could further confirm the applicability of the holographic flow network. Furthermore, since the EHM is actually also valid in the quantum case, an extension of the model beyond the classical limit might be possible. This could reveal unknown relationships between theories, which are connected by a duality, and give us a different perspective in understanding the holographic principle.

Bibliography

- [1] M. S. Albergo, G. Kanwar, and P. E. Shanahan. “Flow-based generative models for Markov chain Monte Carlo in lattice field theory”. In: *Phys. Rev. D* 100 (2019). DOI: 10.1103/PhysRevD.100.034515. arXiv: 1904.12072 [hep-lat].
- [2] M. S. Albergo et al. “Flow-based sampling for fermionic lattice field theories”. In: *Phys. Rev. D* 104 (2021). DOI: 10.1103/PhysRevD.104.114507. arXiv: 2106.05934 [hep-lat].
- [3] V. Balasubramanian, M. Guica, and A. Lawrence. “Holographic interpretations of the renormalization group”. In: *J. High Energ. Phys.* 115 (2013). DOI: 10.1007/JHEP01(2013)115. arXiv: 1211.1729 [hep-th].
- [4] V. Balasubramanian et al. “Holographic Probes of Anti-de Sitter Spacetimes”. In: *Phys. Rev. D* 59 (1998). DOI: 10.1103/PhysRevD.59.104021. arXiv: hep-th/9808017 [hep-th].
- [5] N. Bao et al. “Consistency Conditions for an AdS/MERA Correspondence”. In: *Phys. Rev. D* 97 (2015). DOI: 10.1103/PhysRevD.97.125036. arXiv: 1504.06632 [hep-th].
- [6] C. A. B. Bayona and N. R. F. Braga. “Anti-de Sitter boundary in Poincare coordinates”. In: *Gen. Rel. Grav.* 39 (2005). DOI: 10.1007/s10714-007-0446-y. arXiv: hep-th/0512182 [hep-th].
- [7] J. D. Bekenstein. “Black Holes and Entropy”. In: *Phys. Rev. D* 7 (8 1973), pp. 2333–2346. DOI: 10.1103/PhysRevD.7.2333.
- [8] J. de Boer, E. Verlinde, and H. Verlinde. “On the Holographic Renormalization Group”. In: *JHEP* 8.3 (1999). DOI: 10.1088/1126-6708/2000/08/003. arXiv: hep-th/9912012 [hep-th].
- [9] M. M. Bronstein et al. *Geometric Deep Learning: Grids, Groups, Graphs, Geodesics, and Gauges*. 2021. arXiv: 2104.13478 [cs.LG].

- [10] E. D'Hoker and D. Z. Freedman. "Supersymmetric Gauge Theories and the AdS/CFT Correspondence". In: *Strings, Branes and Extra Dimensions*. 2002. arXiv: [hep-th/0201253](#) [hep-th].
- [11] G. Evenbly and G. Vidal. "Tensor network states and geometry". In: *J. Stat. Phys.* 145 (2011). DOI: [10.1007/s10955-011-0237-4](#). arXiv: [1106.1082](#) [quant-ph].
- [12] M. Finzi, M. Welling, and A. G. Wilson. "A Practical Method for Constructing Equivariant Multilayer Perceptrons for Arbitrary Matrix Groups". In: *Proceedings of Machine Learning Research* 137 (2021). arXiv: [2104.09459](#) [cs.LG]. URL: <https://github.com/mfinzi/equivariant-MLP>.
- [13] M. Fukuma, S. Matsuura, and T. Sakai. "Holographic Renormalization Group". In: *Prog. Theor. Phys.* 109 (2002). DOI: [10.1143/PTP.109.489](#). arXiv: [hep-th/0212314](#) [hep-th].
- [14] K. Hashimoto. "AdS/CFT as a deep Boltzmann machine". In: *Phys. Rev. D* 99 (2019). DOI: [10.1103/PhysRevD.99.106017](#). arXiv: [1903.04951](#) [hep-th].
- [15] K. Hashimoto et al. "Deep Learning and AdS/CFT". In: *Phys. Rev. D* 98 (2018). DOI: [10.1103/PhysRevD.98.046019](#). arXiv: [1802.08313](#) [hep-th].
- [16] K. Hashimoto et al. "Deep Learning and Holographic QCD". In: *Phys. Rev. D* 98 (2018). DOI: [10.1103/PhysRevD.98.106014](#). arXiv: [1809.10536](#) [hep-th]. URL: https://github.com/AkinoriTanaka-physics/DL_holographicQCD.
- [17] I. Heemskerk and J. Polchinski. "Holographic and Wilsonian Renormalization Groups". In: *J. High. Energ. Phys* 31 (2011). DOI: [10.1007/JHEP06\(2011\)031](#). arXiv: [1010.1264](#) [hep-th].
- [18] H.-Y. Hu et al. "Machine Learning Holographic Mapping by Neural Network Renormalization Group". In: *Phys. Rev. Research* 2 (2019). DOI: [10.1103/PhysRevResearch.2.023369](#). arXiv: [1903.00804](#) [cond-mat.dis-nn].
- [19] H.-Y. Hu et al. "RG-Flow: A hierarchical and explainable flow model based on renormalization group and sparse prior". In: *Mach. Learn.: Sci. Technol.* 3 (2020). DOI: [10.1088/2632-2153/ac8393](#). arXiv: [2010.00029](#) [cs.LG]. URL: <https://github.com/hongyehu/RG-Flow>.
- [20] C. H. Lee and X.-L. Qi. "Exact holographic mapping in free fermion systems". In: *Phys. Rev. B* 93 (2015). DOI: [10.1103/PhysRevB.93.035112](#). arXiv: [1503.08592](#) [hep-th].

- [21] S.-H. Li and L. Wang. “Neural Network Renormalization Group”. In: *Phys. Rev. Lett.* 121 (2018). DOI: 10.1103/PhysRevLett.121.260601. arXiv: 1802.02840 [cond-mat.stat-mech]. URL: <https://github.com/li012589/NeuralRG>.
- [22] J. M. Maldacena. “The Large N Limit of Superconformal Field Theories and Supergravity”. In: *Adv. Theor. Math. Phys.* 2 (1997). DOI: 10.1023/A:1026654312961. arXiv: hep-th/9711200 [hep-th].
- [23] G. Papamakarios et al. “Normalizing Flows for Probabilistic Modeling and Inference”. In: *Journal of Machine Learning Research* 22 (2019). arXiv: 1912.02762 [stat.ML].
- [24] X.-L. Qi. “Exact holographic mapping and emergent space-time geometry”. In: *arXiv* (2013). arXiv: 1309.6282 [hep-th].
- [25] D. J. Rezende and S. Mohamed. “Variational Inference with Normalizing Flows”. In: *Proceedings of Machine Learning Research* 37 (2015). arXiv: 1505.05770 [stat.ML].
- [26] L. Susskind and E. Witten. “The Holographic Bound in Anti-de Sitter Space”. In: (1998). arXiv: hep-th/9805114 [hep-th].
- [27] G. Vidal. “A class of quantum many-body states that can be efficiently simulated”. In: *Phys. Rev. Lett.* 101 (2006). DOI: 10.1103/PhysRevLett.101.110501. arXiv: quant-ph/0610099 [quant-ph].
- [28] E. Witten. “Anti De Sitter Space And Holography”. In: *Adv. Theor. Math. Phys.* 2 (1998). arXiv: hep-th/9802150 [hep-th].
- [29] Y.-Z. You, Z. Yang, and X.-L. Qi. “Machine Learning Spatial Geometry from Entanglement Features”. In: *Phys. Rev. B* 97 (2017). DOI: 10.1103/PhysRevB.97.045153. arXiv: 1709.01223 [cond-mat.dis-nn].

A Code Example

The code can be found on GitLab under:

<https://gitlab.physik.uni-muenchen.de/Sven.Krippendorf/geometric-dl-holography>

To train a model, one can run the file `main.py` with python. It will train a model and produce the plots of the two-point function and the holographic distances. Unless parameters and options are specified by flags, the default values in `args.py` are used.

As an example we could choose:

- unitary map (Cayley) as decimator
- linear map as disentangler
- train on GPU

For this setup, we can run the following command:

```
python main.py --decimator su_cayley --disentangler linear --cuda 0
```